Riverdale Public Utility District

Preliminary Engineering Report Wastewater Treatment Plant Improvement Project

Fresno County, CA
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Abbreviations

ADF .... Average Daily Flow
BOD .... Biochemical Oxygen Demand
CEQA .... California Environmental Quality Act
CWSRF .... Clean Water State Revolving Fund
DO .... Dissolved Oxygen
DWR .... Department of Water Resources
EC .... Electrical Conductivity
EDU .... Equivalent Dwelling Unit
fps .... feet per second
gpd .... gallons per day
gpm .... gallons per minute
hp .... horsepower
MBR .... Membrane Bioreactor
MG .... Million Gallons
MGD .... Million Gallons per Day
mg/L .... milligrams per liter
MHI .... Median Household Income
PHF .... Peak Hourly Flow
RPUD .... Riverdale Public Utility District
RWQCB .... Regional Water Quality Control Board
SCADA .... Supervisory Control and Data Acquisition
SWRCB .... State Water Resources Control Board
TSS .... Total Suspended Solids
WDR .... Waste Discharge Requirements
WWTP .... Wastewater Treatment Plant

1 Introduction

1.1 Purpose and Goals

The Riverdale Public Utility District (RPUD) wastewater treatment plant (WWTP) has been operating near capacity for several years and is in need of expansion. The RPUD has obtained Clean Water State Revolving Fund (CWSRF) and Proposition 1 funding to prepare a Preliminary Engineering Report and subsequently prepare the necessary permitting and regulatory compliance documents and the preliminary design of the expanded facilities. The Board of the RPUD directed Provost & Pritchard to proceed with the Preliminary Engineering Report for expansion and upgrade of the WWTP at their April 2016 Board Meeting.

The RPUD WWTP currently operates under a permit issued by the Central Valley Regional Water Quality Control Board (RWQCB), Waste Discharge Requirements (WDR) Order No. 85-252. The permitted capacity of the existing pond-based WWTP is 0.25 million gallons per day (MGD). The facility presently operates at approximately 0.22 MGD (average daily flow), or 88% of the treatment capacity. The RWQCB typically requires planning for capacity expansion to occur when a system reaches 80% of the design capacity.

The objectives of the proposed improvement projects are to provide the necessary capacity expansion and treatment process upgrades to provide more reliable biological treatment. The purpose of this document is to address the necessary design capacity and WWTP performance criteria, and evaluate and develop recommendations related to the treatment process and disposal method. To accommodate projected growth and development, the WWTP will be expanded to an average daily design flow rate of 0.325 MGD. The design criteria for influent wastewater characteristics, flow and effluent requirements are summarized in Table 1-1.

Table 1-1 Wastewater Characteristics and Treatment Criteria

Design Criteria	Existing Design Value	Proposed Design Value
Design Flow		
Average Daily Flow (ADF)	0.25 MGD	0.325 MGD
Maximum Daily Flow (Peaking Factor = 1.5 x ADF)	0.375 MGD	0.488 MGD
Peak Hourly Flow (Peaking Factor = 3.7 x ADF)	0.75 MGD	1.20 MGD
Influent Wastewater Characteristics*		
Biochemical Oxygen Demand (BOD_5)	200 mg/L	250 mg/L
Suspended Solids	200 mg/L	250 mg/L
Design Discharge Requirements		
Biochemical Oxygen Demand (BOD_5) (30-day mean/maximum)	40/80 mg/L	30/60 mg/L
Suspended Solids (30-day mean/maximum)	40/80 mg/L	30/60 mg/L
Total Nitrogen	N/A	No Limit
EC**	N/A	500 + source
Effluent pH	6.5-8.5	6.5-8.5
Dissolved Oxygen	1.0 mg/L	1.0 mg/L

* Proposed influent wastewater design values for BOD and TSS concentrations are increased from the existing design values due to water conservation efforts and water efficient features on new development that will result in lower unit flows and thus higher loading concentrations.

**Source water EC is very high (approx 1,100 µmhos/cm). It is anticipated that the discharge limit will be based on the source water quality plus 500 µmhos/cm.

2 Project Area

2.1 Location

The RPUD is located approximately 25 miles south of the City of Fresno, in rural Fresno County, California. A vicinity map is included as Figure 2-1. Riverdale is a predominantly agricultural community with no large industrial businesses present. The community had a population of 3,153 in the year 2010, according to US Census data. The current population is estimated to be 3,197 based on the current number of residential sewer service connections (864) and approximately 3.7 persons per household. Figure 2-2 shows a map of the service area.

The topography in Riverdale is flat, with typical ground slopes of less than 0.2 percent. The estimated ground elevation is 215 feet at the WWTP site, based on the Riverdale USGS Quadrangle Map (Figure 2-3). Groundwater elevations were approximately 30 feet (elevation) in the fall of 2015, based on information obtained from the Department of Water Resources (DWR), as shown in Figure 2-4 groundwater contour map. Groundwater elevations have gradually declined since the spring of 2000, when the groundwater elevation was reported to be about 110 feet (elevation). DWR groundwater contour maps for the spring of 2016 have shown even further decline to 0-10 feet elevation, showing a decline in groundwater levels of about 100 feet over the past 16 years..

The WWTP is located east of the Brawley Avenue alignment and south of Cerini Avenue, approximately 1.5 miles north of the Riverdale community. The existing WWTP site is on APN 053-090-011, located in Section 13, Township 17 South, Range 19 East, MDB&M, at Latitude 36° 27' 25" North, Longitude 119° 52' 00" West. An additional 40-acre tract of land was acquired by the District, which adjoins the west boundary of the existing property and consists of the Southwest ¼ of the Northwest ¼ of Section 13. The Van Ness Slough remnant is located in the southeast corner of this additional 40-acre property. The Van Ness Slough remnant will be avoided, limiting the 40-acre site to approximately 33 acres of useable land for wastewater disposal.

2.2 Current Land Use and Land Use Trends

The land use around Riverdale is primarily agricultural. The large majority of the land use within the RPUD service area is residential, with some commercial, institutional and limited industrial. Figure 2-5 shows the land uses in and around the RPUD service area.

2.3 Current System Users

The RPUD currently serves approximately 923 sewer services, including 864 residential services, 47 commercial businesses, 9 churches and libraries, and 3 schools. There are no significant industrial users served by the RPUD WWTP.

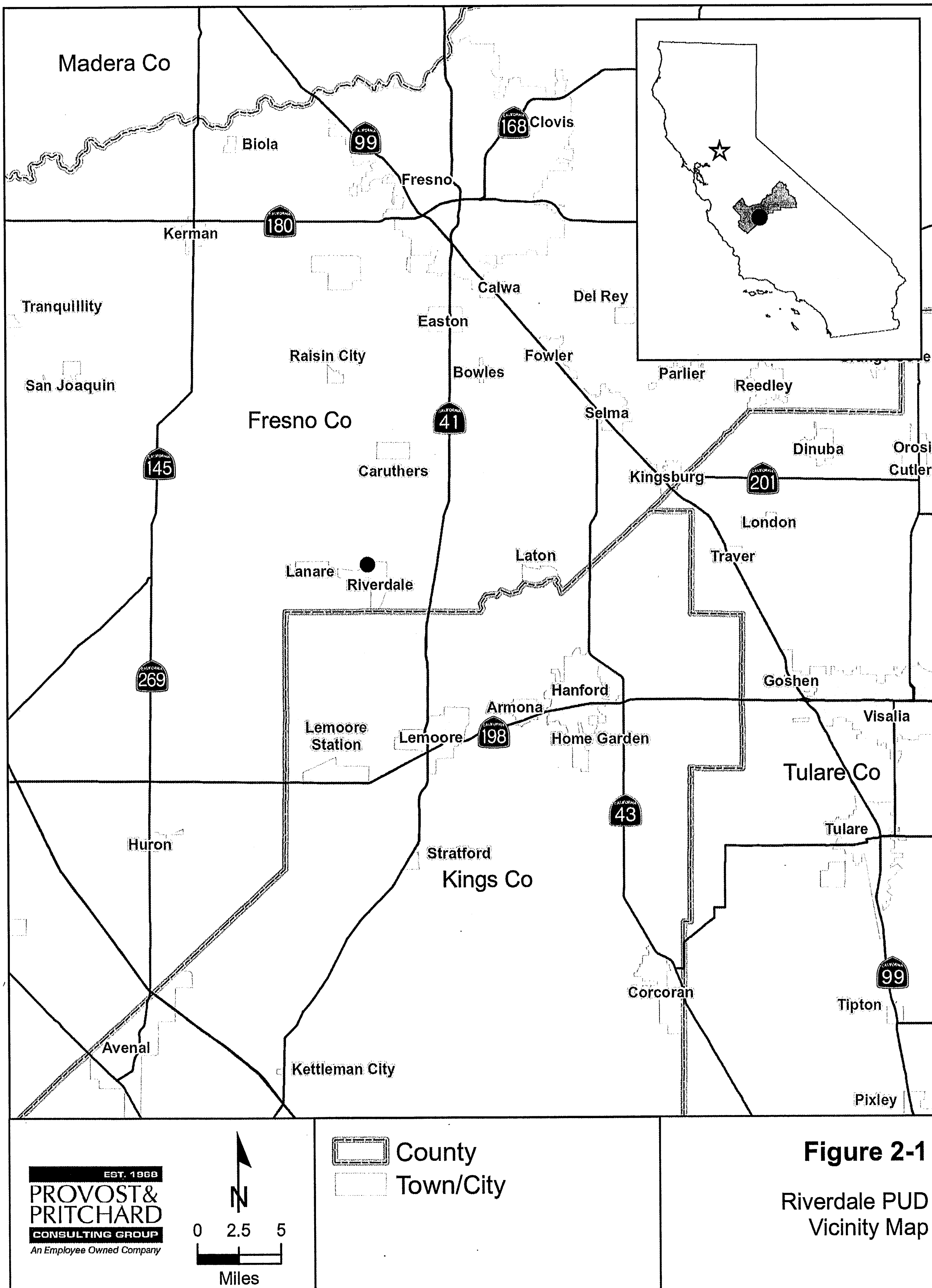

Figure 2-1

Riverdale PUD Vicinity Map

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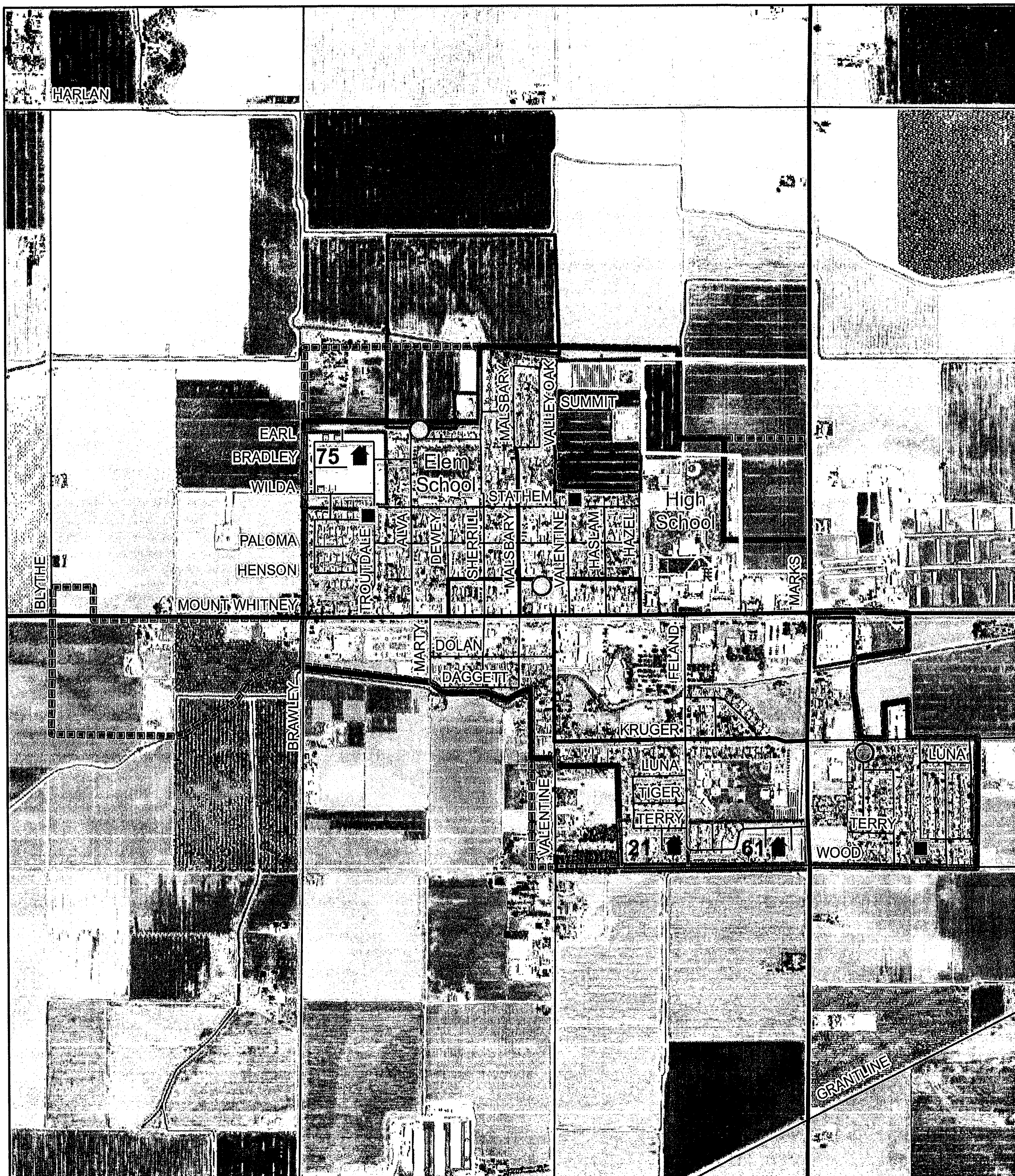

EST. 1968 PROVOST & PRITCHARD CONSULTING GROUP
An Employee Owned Company

N

0 400 800
Feet

Wells
Lift pumps
Minor Road
Major Road
School
New development
PUD Boundary
SOI Boundary
Proposed development

Figure 2-2

Riverdale PUD Service Area

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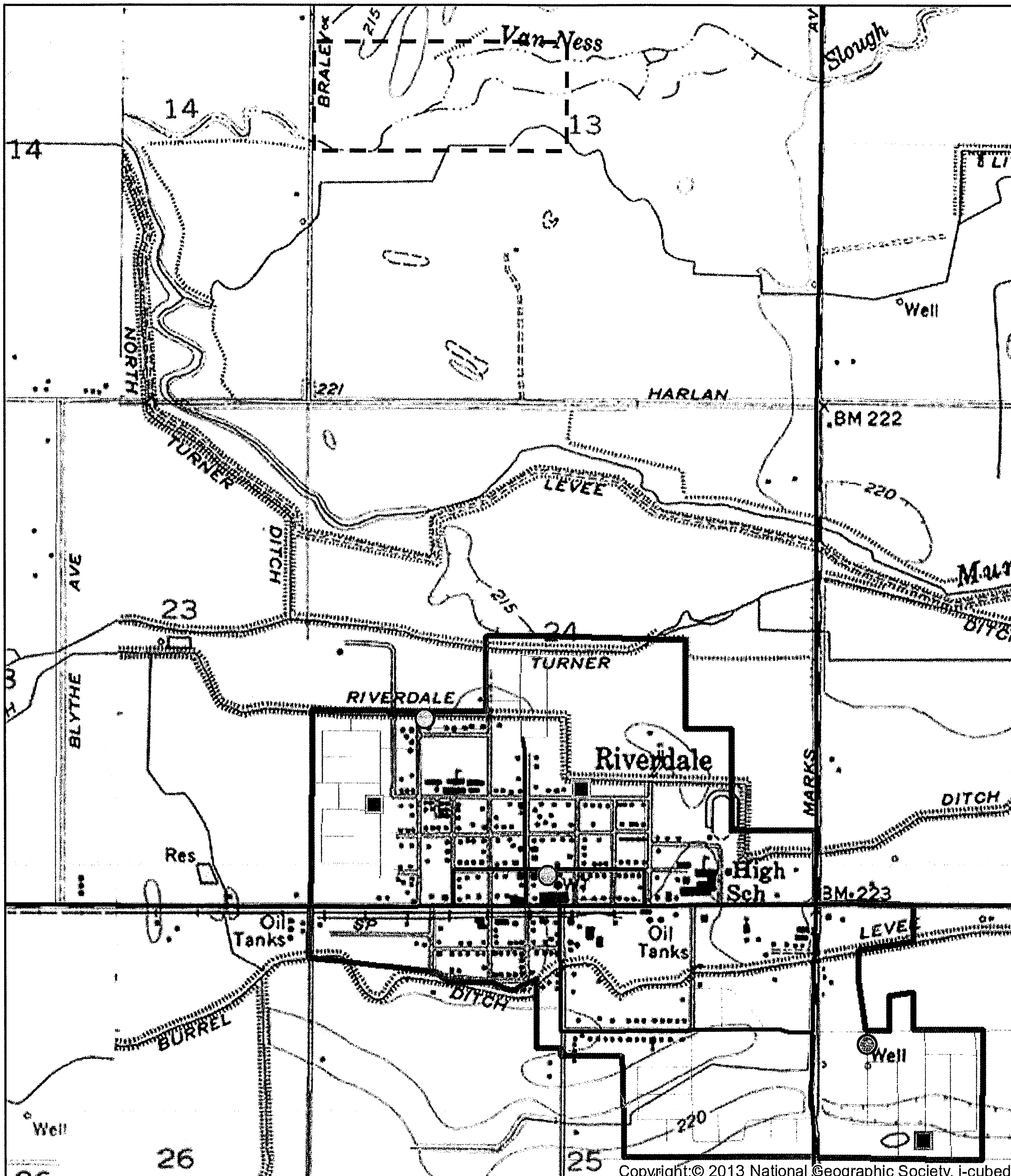

EST. 1968
PROVOST& PRITCHARD
CONSULTING GROUP
An Employee Owned Company

N

0 400 800
Feet

- Wastewater Treatment Plant
- Well
- Lift pump
- Minor Road
- Major Road
- PUD Boundary

Figure 2-3
Riverdale PUD
USGS Topo
Quadrangle Map

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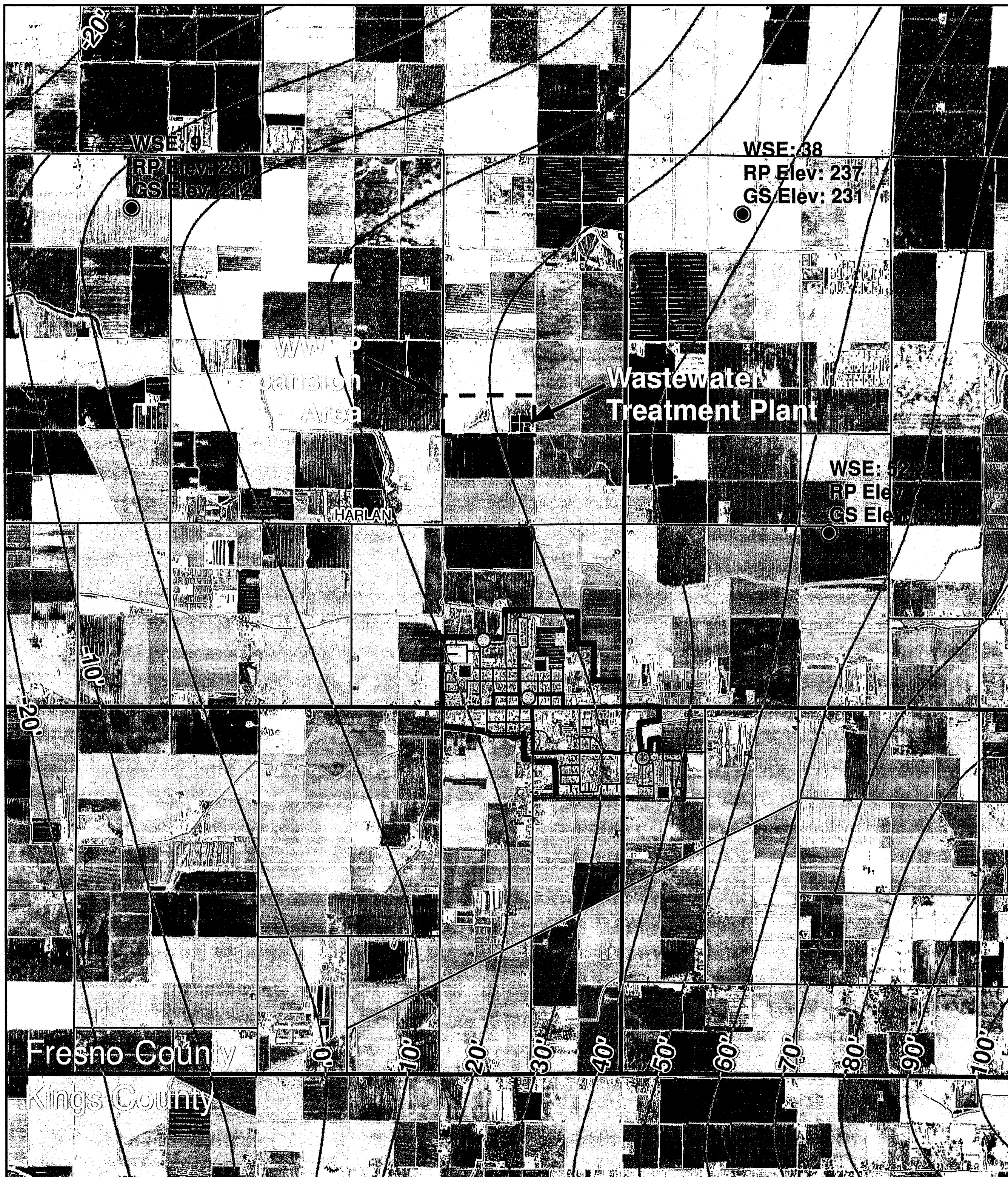

EST. 1968
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PRITCHARD
CONSULTING GROUP
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0 0.25 0.5
Miles

- Wells
- Lift pumps
- Fall 2015 Well Point (DWR GIC)
- Fall 2015 WSE Contour (DWR GIC)
- Minor Road
- Major Road
- PUD Boundary

GS = Ground Surface
RP = Reference Point
WSE = Water Surface Elevation
DWR = Dept. of Water Resources
GIC = Groundwater Information Center

Figure 2-4
Groundwater
Contour Map

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Figure 2-5

Riverdale PUD Land Use Map

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2.4 Growth Areas and Population Trends

Census population data for the years 2000 and 2010 indicate the population increased from 2,416 to 3,153. The population growth rate for this period was therefore approximately 2.7% per year. By contrast, a review of the District's residential sewer connection list in 2005 and 2016 showed that the residential service connections increased from 809 to 864 during that period. The growth rate for this period was less than 1% per year.

Based on the State of California, Department of Finance, Report P-1 (Total Population): State and County Population Projections, 2010-2060 (December 2014), the estimated average growth rate for Fresno County over the next 20 years is expected to be approximately 1.5%.

Based on historic growth rates for Riverdale, and understanding that growth rates will fluctuate over any 20-year period, it is recommended that a growth rate of 1.5% to 2% per year be used for projecting future growth. At this rate of growth rate, and based on the current 923 total service connections, the projected number of sewer service connections in 2035 would be approximately 1,243 to 1,372, or an increase of 320 to 449 connections. This would equate to a population of about 4,600 to 5,080.

The anticipated actual project development in Riverdale should also be a considered. Figure 2-6 shows the locations of the anticipated development projects within the District, and Table 2-1, on the following page, shows the current will-serve commitments. It is reasonable to assume that this proposed development will occur within the 20-year planning period. The proposed development would add approximately 383 homes, which is within the 20-year growth range discussed above. This suggests that the estimated growth rate for the District is realistic for facility planning purposes.

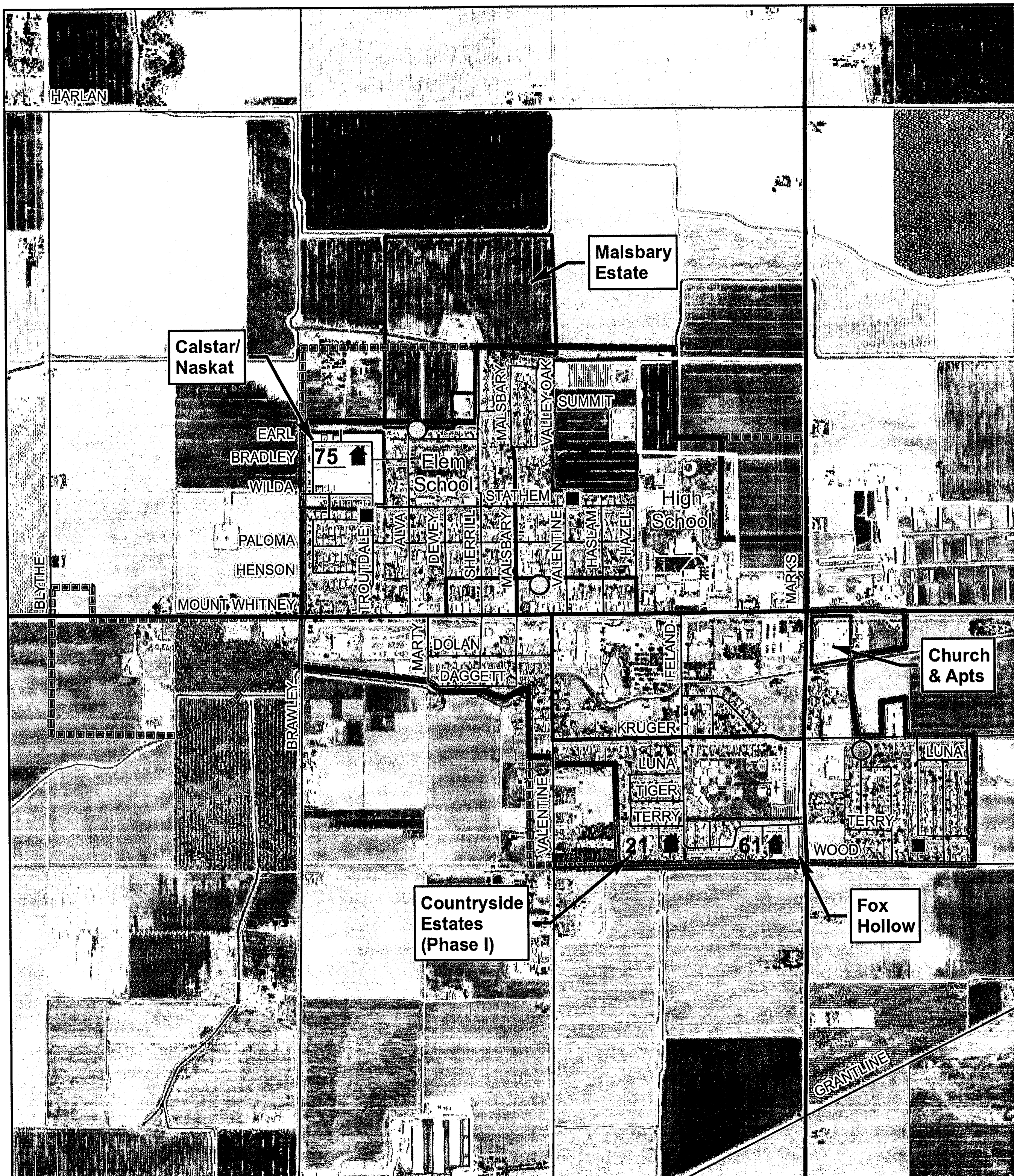
Table 2-1 Riverdale PUD Will-Serve Commitments

Development Project/Demand	Demand		Remaining Capacity		Notes
	(EDU)[(2)]	*(gpd)*	*(gpd)*	*(EDU)*[(2)]	
Existing System Capacity			**250,000**		
Capacity used	*923*[(3)]	*220,000*[(1)]	*30,000*	136	Assumed EDUs
WWTP Expansion (0.325 mgd)	**341**	**75,000**	**105,000**	**477**	
Jonathan Homes - Fox Hollow (Proposed Units)	26	5,720	99,280	451	Previously Countryside Estates Phase 2. 35 of 61 lots have been built. 26 remain vacant.
CalStar/NASKAT (Proposed Units)	57	12,540	86,740	394	18 of 75 lots have been built. 57 remain vacant.
Malsbary Estate *(Phase 1)* (Proposed Units)	69	15,180	71,560	325	Inside District SOI
Malsbary Estate *(Future Phases)* (Proposed Units)	231	50,820	20,740	94	Adjacent to District SOI
Total	**1,306**	**304,260**	**20,740**	**94**	

Assumptions:			
Persons per unit (Census 2010 Riverdale CDP):	3.7		
Sewer load for new development[(2)]:	60 gpcd	220	gpd/EDU

Notes:

1) Existing capacity used is based on WWTP flow data.
2) Capacity projections for future growth are based on recommended design values for new development.
3) 2016 Residential Customers: 864 sewer; 2016 Total Customers: 923 sewer.



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N

0 400 800 Feet

Wells
Lift pumps
Minor Road
Major Road
School
New development
PUD Boundary
SOI Boundary
Proposed development

Figure 2-6

Proposed Development

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3 Existing Facilities

3.1 Existing Facilities

The existing Riverdale PUD WWTP is an aerated lagoon system designed to remove biochemical oxygen demand (BOD) and suspended solids from the community's wastewater. The raw influent is received at the headworks structure, where it passes through a comminutor before going into the complete-mix (CM) aerated lagoon. Effluent from the CM lagoon is distributed to the Oxidation Lagoons, which may be operated in series or in parallel. The Oxidation Lagoons provide for percolation of the effluent into the soil and evaporation to the atmosphere. Effluent from the lagoons may also be applied to the pasture lands adjacent to the WWTP. The existing plant layout is shown in Figure 3-1.

3.1.1 Influent Pumping, Force Main and Headworks

Sewage is collected from throughout the community in a network of gravity sewers which come together at a lift station located at the corner of Valentine and Stathem Avenues. A new lift station has been constructed next to the old lift station. The new lift station is equipped with two 420 gpm submersible pumps, with room for a third pump. Each pump was sized to handle the existing peak hour flow to provide redundancy. The lead pump alternates at each startup to equally exercise each pump and minimize the number of starts per hour for each pump. The pumps operate on a variable frequency drive. The pump speed gradually increases at startup, runs at a constant maximum speed at the level set point. The lift station is integrated into the RPUD SCADA system centrally located at the District Office.

All sewage from the community passes through this lift station, which pumps it to the WWTP approximately 1.5 miles north of the community through a 10-inch sewer force main. The existing 10-inch sewer main to the WWTP was installed in 1958. The condition of the sewer force main is unknown. Air release valves that were installed with the original force main construction have been removed.

Influent flow is measured at the WWTP by an ultrasonic flow meter that was installed in July, 2002. The influent then enters a headworks structure, where it passes through a comminutor before passing into a complete-mix (CM) aerated lagoon. The Muffin Monster comminutor (model 30005-0018) manufactured by JWC, Inc. is rated at a maximum capacity of 725 gpm. There is also a bypass channel with a manual bar screen.

3.1.2 Treatment Ponds

The existing WWTP includes a CM lagoon with two (2) surface aerators, and six (6) non-aerated oxidation lagoons. The volumes, surface area and depth of the existing ponds are summarized in Table 3-1.

The CM lagoon is designed to maintain all settleable solids in suspension and to provide an environment for development of a biological floc. The incoming wastewater provides organic food for continued growth and maintenance of biological activity, which breaks down organic material and results in the reduction of the biochemical oxygen demand (BOD).

The oxidation lagoons are designed to degrade organic contaminants by aerobic and facultative bacteria, reduce the BOD, and settle out suspended solids.

The CM lagoon is concrete lined and was constructed in 1984. During construction, the condition of the liner will be inspected and can be considered for use as an emergency bypass of the proposed treatment facilities.

The oxidation lagoons are not lined. Oxidation lagoons 1 through 3 were constructed in 1958. Oxidation lagoons 4 through 6 were constructed in 1984. The condition of the lagoons is unknown, but the oxidation lagoons that were constructed in 1984 are assumed to be acceptable with some maintenance and sludge removal. These will be inspected during construction of the proposed improvements. The other existing oxidation lagoons are old and would likely require sludge removal, reconstruction of the slopes, replacement of piping, and other maintenance. Rather than refurbishing these ponds, they will be filled to allow space for treatment equipment.

Table 3-1 Summary of Existing Treatment Ponds

Pond	Volume MG	Surface Area Acre	Water Depth ft
CM Lagoon	0.50	0.21	10
Oxidation Lagoon 1	0.70	0.54	4
Oxidation Lagoon 2	0.70	0.54	4
Oxidation Lagoon 3	1.98	1.52	4
Oxidation Lagoon 4	0.78	0.60	4
Oxidation Lagoon 5	1.70	1.32	4
Oxidation Lagoon 6	1.41	1.09	4
Oxidation Lagoon Total	*7.27*	*5.61*	
Total	7.77	5.82	

3.1.3 Effluent Disposal

Evaporation and percolation occur in the existing oxidation lagoons. Effluent from the oxidation lagoons is released to the adjacent reclamation area for disposal, as needed.

3.1.4 Electrical Facilities

The existing electrical facilities include a meter panel and PG&E service with 240 Volt, 3 phase, 4 wire service. The electrical control panels provide for monitoring and control of the flow meter, aerators and comminutor.

The existing electrical facilities are old and likely in need of replacement.

3.2 Sources of Wastewater

Wastewater that is delivered to the WWTP is from the community of Riverdale, and consists primarily of residential flows, with some commercial uses. There are no industrial sources delivered to the WWTP.

3.3 Discharge Violations

The Riverdale PUD WWTP has been cited for multiple violations of its Discharge order, for BOD, DO, and EC. There are no current enforcement actions, although the District received Notices of Violation on 4/30/2002, 6/18/2003, 12/31/2003, and 1/5/2006. Appendix A includes a list of violations that have occurred over the past five years.

3.4 WWTP Influent Characteristics

Influent flows to the WWTP are primarily from residential uses, with some commercial and institutional connections. Influent flow characteristics are summarized in Table 3-2. The sampling results for influent BOD and TSS are very low and not considered to be appropriate for design considerations based on typical residential raw wastewater characteristics. Previous design criteria for the WWTP utilized influent BOD and TSS concentrations of 200 mg/L. Due to water conservation efforts and Building Code requirements for water efficient plumbing on new construction, it is anticipated that per capita flows will go down, causing BOD and TSS concentrations to increase. The design criteria for wastewater influent for the current proposed WWTP improvements therefore used BOD and TSS concentrations of 250 mg/L each.

Table 3-2 Influent Flow Characteristics

		Average	5/25/2016			5/26/2016		
			7:00 AM	12:00 PM	2:30 PM	6:00 AM	11:00 AM	2:00 PM
Alkalinity as CaCO3	mg/L	427	410	410	440	430	440	430
Bicarbonate as CaCO3	mg/L	427	410	410	440	430	440	430
Carbonate as CaCO3	mg/L	ND	ND	ND	ND	ND	ND	ND
Hydroxide as CaCO3	mg/L	ND	ND	ND	ND	ND	ND	ND
BOD	mg/L	106	90	71	120	100	170	82
pH		7.9	8.0	8.0	8.0	7.9	7.7	8.0
Settleable Solids	mL/L	2.5	1.9	1.8	3.6	0.9	5.3	1.2
Total Kjeldahl Nitrogen	mg/L	45	42	43	51	42	49	42
TSS	mg/L	130	130	140	110	120	160	120

3.5 WWTP Effluent Characteristics

The existing WWTP effluent is secondary treated effluent without denitrification or disinfection. BOD levels fluctuate considerably, and are typically above the WDR limit of 40 mg/L (30-day mean) and 80 mg/L (maximum). The average effluent BOD from February 2014 through April 2016 was nearly 100 mg/L, with a minimum level of 28 mg/L in September 2015, followed by a maximum level of 320 mg/L in October 2015.

Effluent EC levels are also high, which is due primarily to the source water EC being approximately 1,100 to 1,200 µmhos/cm. Effluent EC levels averaged approximately 1,540 µmhos/cm from July 2015 through April 2016, since the new source Well 6 has been online. Well 6 produces water with higher levels of EC than the existing wells that it replaced.

Effluent data is provided in Table 3-3.

Table 3-3 Effluent Quality Data

Date	Pond D.O.	E.C.	BOD	S.S.
2/5/2014	3.9	919	74	ND
2/13/2014	7.9	964		
2/19/2014	8	945		ND
2/26/2014	7.4	999		
3/5/2014	9.6	No Read	100	ND
3/12/2014	0.8	No Read		
3/19/2014	4.9	No Read		ND
3/26/2014	3.7	No Read		
4/2/2014	3	1034		
4/9/2014	2.3	1270	94	ND
4/16/2014	3.8	1349		
4/23/2014	4.3	1159		ND
4/30/2014	6.4	1223		
5/7/2014	13.8	No Read	45	ND
5/14/2014	2.6	No Read		
5/21/2014	1.5	No Read		ND
5/28/2014	2.7	No Read		
6/4/2014	2.2	No Read	61	ND
6/12/2014	2.4	No Read		
6/18/2014	5.6	No Read		ND
6/25/2014	6.5	No Read		
7/2/2014	8	1175		ND
7/9/2014	1.3	1561	58	
7/16/2014	2.8	No Read		
7/23/2014	1.9	No Read		ND
7/30/2014	0.9	No Read		
8/6/2014	1.3	1121	96	ND
8/13/2014	1.6	1249		
8/20/2014	3.6	1369		ND
8/27/2014	1.5	1417		
9/3/2014	3.6	1514	76	ND
9/10/2014	0.3	1475		
9/17/2014	0.7	1390		ND
9/25/2014	2.3	1593		
10/5/2014	0.9	1476		

Date	Pond D.O.	E.C.	BOD	S.S.
10/8/2014	1.5	1385	140	ND
10/22/2014	1	1396		ND
10/29/2014	1.2	1324		
11/5/2014	3.6	1265	69	ND
11/12/2014	2.6	1332		
11/19/2014	3	1296		ND
11/26/2014	4.9	1211		
12/3/2014	3.2	1117	87	ND
12/11/2014	1.8	1127		
12/17/2014	2.6	1223		ND
12/24/2014	1.9	1191		
1/7/2015	13.8	1293	98	ND
1/14/2015	4.4	1129		
1/21/2015	1.1	1116		ND
3/4/2015	4.4	1191	130	ND
3/12/2015	1.7	1231		
3/18/2015	1.1	1324		ND
3/25/2015	1.7	1333		
6/4/2015	2.9	1583	120	ND
6/10/2015	2.6	1711		
6/17/2015	2.4	1772		ND
6/24/2015	1.7	1744		
7/1/2015	2.9	1813	56	ND
7/1/2015	2.4	1813	56	ND
7/8/2015	5.4	1910		
7/8/2015	5.4	1910		
7/16/2015	4.2	2010		ND
7/16/2015	4.2	2010		ND
7/22/2015	3.8	1860		
7/27/2015	3.8	1860		
8/3/2015	2.7	1743	82	ND
8/5/2015	2.7	1743	82	ND
8/12/2015	3	1735		
8/12/2015	3	1735		ND
8/19/2015	0.8	1819		ND
8/19/2015	0.8	1819		

Date	Pond D.O.	E.C.	BOD	S.S.
8/26/2015	1.8	1805		
8/26/2015	1.8	1748		
9/2/2015	1.4	1748		ND
9/9/2015	5.3	1746	28	
9/16/2015	1.2	1638		ND
9/23/2015	1.3	1662		
9/30/2015	1.3	1678		
10/7/2015	2.2	1209	320	ND
10/14/2015	2.1	1698		
10/21/2015	2.3	1566		ND
10/22/2015	1.9	1518		
11/4/2015	2.8	1370	93	ND
11/11/2015	3.2	1362		
11/18/2015	3.6	1225		ND
11/24/2015	3	1260		
12/2/2015	2.7	1145	180	ND
12/9/2015	2.6	1250		
12/16/2015	1.9	1214		ND
12/23/2015	1.5	1179		
2/3/2016	5.6	1016	62	ND
2/10/2016	5	1027		
2/17/2016	3.6	1288		ND
2/24/2016	2.8	1166		
4/6/2016	2	1364	180	ND
4/14/2016	3.6	1390		
4/20/2016	5	1220		ND
4/27/2016	3.8	1376		
Average	**3.3**	**1423**	**99**	
Minimum	0.3	919	28	
Maximum	13.8	2010	320	

4 Need for Project

The existing WWTP has been operating at capacity for several years, and is need of expansion. The existing WWTP has also been unable to achieve the required BOD removal. The proposed upgrade of the existing WWTP includes both a capacity expansion and treatment process upgrade to provide more reliable biological treatment.

Further treatment of the effluent to disinfected tertiary quality could increase options for types of crops that could use the effluent for irrigation. However, since there is more than sufficient percolation and potential fodder crop acreage available to dispose the current and planned flows at the secondary treatment level, the effort to pursue more reclamation options and the ongoing cost to operate the higher-level treatment plant has been determined to not be cost effective at this time.

4.1 Capacity Evaluation

4.1.1 Existing WWTP Flows

The existing WWTP capacity is 250,000 gpd. A Preliminary Design Report prepared in 2006 proposed a WWTP expansion to increase capacity to 424,000 gpd, expandable to an ultimate future capacity of 570,000 gpd. Growth within the District has not occurred at the rates projected in the 2006 Report, since changes in the economy caused a slowdown in development. In fact, the WWTP has experienced a decline in influent flows since peaking in 2012.

In 2004, the actual WWTP annual average flow rate was 232,000 gpd and the maximum month average was 256,000 gpd. Recent flow data was reviewed for the period between January 2011 and December 2016. The highest annual average flow rate occurred in 2012 at 251,000 gpd, with the maximum month average of 258,000 gpd occurring in February 2012 (see Table 4-1). However, since 2012 there has been a steady decline in flow. Since population has not decreased, the decline in flows is considered to be attributed to water use reductions within the existing households, which is likely the result of the drought and water conservation efforts. It is expected that these declines are not temporary, but that due to ongoing conservation efforts mandated by the State, per capita water use in the existing housing stock will remain at or near this lower level. Assuming that there may be some rebound in per capita use over time, it is recommended that the 2015 maximum month average flow rate of 220,000 gpd be used as the baseline current demand for projecting future flows.

Table 4-1 Riverdale PUD WWTP Average Monthly Influent Flows (in MGD)

Month	2011	2012	2013	2014	2015	2016
January	0.240	0.252	0.245	0.215	0.204	0.208
February	0.235	0.258	0.238	0.214	0.215	0.207
March	0.236	0.250	0.233	0.213	0.213	0.207
April	0.237	0.253	0.227	0.211	0.209	0.202
May	0.233	0.254	0.226	0.207	0.203	0.207
June	0.246	0.243	0.228	0.210	0.204	0.199
July	0.243	0.248	0.227	0.215	0.216	0.188
August	0.246	0.253	0.235	0.221	0.214	0.197

Month	2011	2012	2013	2014	2015	2016
September	0.246	0.253	0.242	0.222	0.215	0.191
October	0.250	0.255	0.241	0.226	0.220	0.198
November	0.252	0.256	0.230	0.222	0.220	0.201
December	0.249	0.241	0.211	0.218	0.209	0.199
Annual Average	**0.243**	**0.251**	**0.232**	**0.216**	**0.212**	**0.200**
Maximum Month[1.]	**0.252**	**0.258**	**0.245**	**0.226**	**0.220**	**0.208**

Note:
1. Maximum month shown above represents the maximum month in the year and not the 30-day average daily "dry" weather flow. Sizing based upon the maximum month provides a degree of conservatism to the flow projections.

4.1.2 Projected Growth

The 2006 Report assumed a 10-year planning period with a growth rate of 50 new homes per year from 2005 through 2014. These assumptions were the basis for the 424,000 gpd design flow rate at that time. The equivalent population for this flow rate would have been 4,442 in 2014. Actual growth was significantly less due to a slowdown in the housing market.

Census population data for the years 2000 and 2010 indicate the population increased from 2,416 to 3,153. The population growth rate for the Riverdale census tract during this period was therefore approximately 2.7% per year. A review of the District's residential sewer connection list in 2005 and 2016 showed that the residential service connections increased from 809 to 864 during that period. The growth rate for the District during this period was less than 1% per year, suggesting that most growth in the Riverdale census tract during the period 2000 through 2010 was outside the District's service area boundaries, or that the majority of growth occurred between 2000 and 2005.

Based on the State of California, Department of Finance, Report P-1 (Total Population): State and County Population Projections, 2010-2060 (December 2014), the estimated average growth rate for Fresno County over the next 20 years is expected to be approximately 1.5%.

Based on historic growth rates for Riverdale, and understanding that growth rates will fluctuate over any 20-year period, it is recommended that a growth rate of 1.5% to 2% per year be used for projecting future WWTP flows. At this rate of growth rate, and based on the current 923 total service connections, the projected number of sewer service connections in 2035 would be approximately 1,243 to 1,372, or an increase of 320 to 449 connections.

The anticipated development in Riverdale should also be a considered. A map of locations of the anticipated development is shown in Figure 2-6. It is reasonable to assume that this proposed development may occur within the 20-year planning period. The proposed development would add approximately 383 homes, which is within the 20-year growth range discussed above. This suggests that the estimated growth rate for the District is realistic for facility planning purposes.

4.2 Projected Future Flows

The WWTP is currently operating near capacity. Typically, the Regional Water Quality Control Board requires planning for capacity expansion to occur at 80% of the design capacity. With current flows nearing 220,000 gpd, the plant is operating at about 88% of its existing design capacity. A typical planning period is 20 years, or the payback period for a loan (if required to fund the expansion), whichever is greater.

Table 4-2 presents estimates of future WWTP flows using the current flow rate observed and projected flow rates in 20 years based on a 1.5% and 2.0% growth rates.

Table 4-2 Projected Future Flows

	Number of Connections	Flow Rate ADF (gpd)	Number of Connections	Flow Rate ADF (gpd)
Planned Growth Rate	**1.5%**		**2.0%**	
Current number of connections	923	220,000	923	220,000
Projected increase in 20 years	320	70,400	449	98,800
Total	1,243	290,400	1,372	318,800
Planned design capacity				**325,000**

Note: ADF flow rate calculation is based on a 220 gpd per each new sewer connection (60 gpcd x 3.7 persons per household).

It is noted that, while current per capita sewer flows average about 66 gallons per capita per day (gpcd), per capita flows are expected to continue to decline due to 2013 Building Code requirements for water efficient appliances and fixtures in new homes, which are intended to result in a 20-percent reduction in interior water use. A 2015 study by ConSol, funded by the California Homebuilding Foundation[1], found that indoor water use decreased over 50 percent between 1975 and 2015, with the current use being less than 130 gallons per household per day, or approximately 32 gallons per capita per day at a household formation rate of 4.0 persons per household. That research places water use at a dramatically lower rate than the industry has used until recent years, which has typically been about 75 gpcd. While the study's use rate may be achieved and proved out by future water use measurements, a more conservative number should be used for planning until that aggressive number is proven by experience. The stated goal of the 2013 CalGreen code is to achieve a 20 percent reduction in indoor water use. Applying that percentage to the older 75 gpcd use rate results in a unit flow of 60 gpcd, which has been used to project flows from new development for this report.

It is recommended that the WWTP expansion provide a capacity of 325,000 gpd (ADF). The selected expansion capacity will be used to design the WWTP improvements and for the new Waste Discharge Permit flow limit with the Regional Water Quality Control Board.

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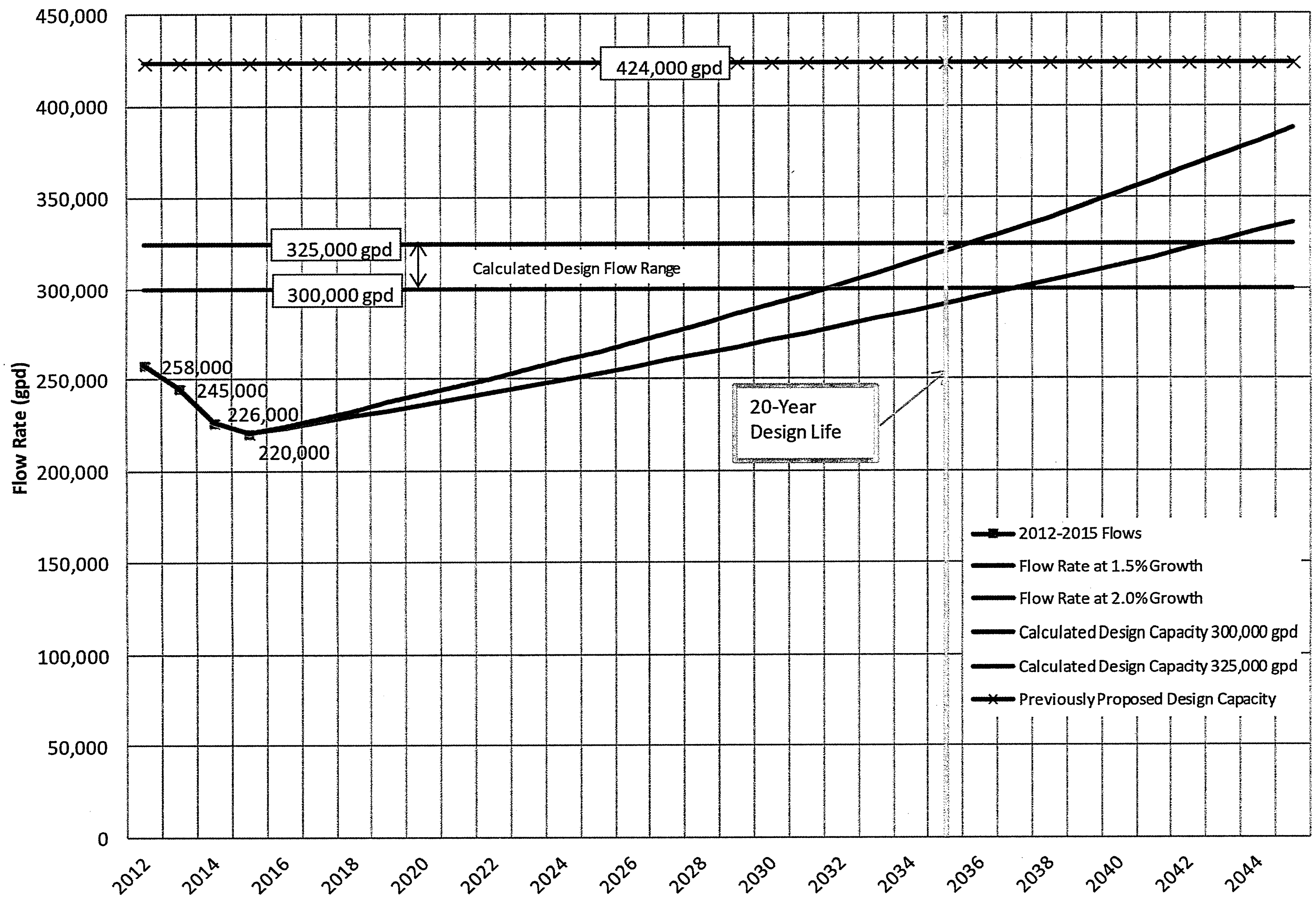
1 "Codes and Standards Research Report: California's Residential Indoor Water Use," 2nd Edition, Revised May 8, 2015, ConSol, Stockton, CA

Figure 4-1 Riverdale PUD Wastewater Flow Projections

Riverdale PUD Wastewater Flow Projections



4.3 Waste Discharge Requirements

The current WDRs were issued in 1985, and new WDRs to be issued with the proposed project are expected to have more restrictive effluent limits and greater testing requirements. The new WDRs should be based on a design flow of 0.325 MGD.

Removal of nitrogen is not a requirement of the current WDRs. Based on a preliminary antidegradation analysis and correspondence with the RWQCB, removal of nitrogen is not anticipated to be a requirement in the WDRs for the proposed treatment improvements.

BOD limits in the current WDRs include a 30-day mean of 40 mg/L and a maximum of 80 mg/L. It is expected that the BOD and TSS limits in the new WDRs will each include a 30-day mean of 30 mg/L and a maximum of 60 mg/L.

Current pH limits are 6.5 to 9.0, and are not expected to change.

The current WDRs state that the maximum EC of the discharge shall not exceed 1,100 µmhos/cm. However, as discussed with the RWQCB at a meeting on October 1, 2015, and acknowledged in a summary letter dated February 10, 2016, the EC level of the new source wells in the Riverdale community is about 1,100 – 1,200 µmhos/cm. The summary letter is attached in Appendix B.

Based on recent correspondence with the RWQCB, RPUD understands that the 1,100 µmhos/cm EC limit will not be imposed in the new WDRs. The new WDRs will instead include a requirement that the effluent EC shall not exceed the source water EC plus 500 µmhos/cm.

5 Project Alternatives Analysis

Four (4) treatment processes (Aeration Ponds, Biolac or Bioworks, STM Aerotor, and Membrane Bioreactor (MBR)) have been evaluated to fulfill the needs for the project as discussed in previous sections. In addition to treatment process alternatives, there are several project components that will be necessary regardless of the selected treatment alternative. The improvements common to all alternatives are also discussed in this section.

Consolidation or regionalization was not considered as an alternative because there are no existing wastewater systems within 5 miles of Riverdale.

5.1 Design Criteria

Design criteria for the proposed WWTP improvements are presented in Table 5-1.

Table 5-1 Basic Criteria of Design

Location/ Process	Unit		
Elevation	ft	**215**	
Influent BOD	mg/L	**250**	
Influent TSS	mg/L	**250**	
Influent Free Ammonia	mg/L	**45**	
Minimum water temperature	deg F	**50**	
Maximum water temperature	deg F	**70**	
Average daily flow	mgd	**0.325**	
	gpm	**226**	
Maximum Day (Peaking Factor = 1.5 x ADF)	mgd	**0.488**	
Peak Hourly Flow (Peaking Factor = 3.7 x ADF)	gpm	**840**	
Minimum hourly flow	gpm	**113**	
Secondary effluent		**Mean**	**Maximum**
BOD_5	mg/L	**30**	**60**
Solids, Total Suspended (TSS)	mg/L	**30**	**60**
Total Nitrogen (TN)	mg/L	**No Limit**	

5.2 Improvements Common to All Alternatives

5.2.1 Sewer Lift Station Upgrades and Force Main Replacement

All sewage from the community passes through the lift station located at the corner of Valentine and Stathem Avenues, which pumps it to the WWTP approximately 1.5 miles north of the community through a 10-inch sewer force main. The existing force main to the WWTP was installed in 1958. The pipe material is asbestoscement and the condition is unknown. According to the District, air valves along the force main have been removed, therefore the existing pipe is susceptible to air accumulating at high points. If only one pump is running (which is the typical operation), the velocities in the force main would be less than 2.0 feet per second, making the pipe susceptible to the buildup of settled solids. The force main is assumed to be past its useful life. The force main is a critical component of the system since it conveys 100 percent of all influent to the WWTP. For these reasons it is recommended that the existing force main be replaced.

A new lift station was recently constructed. This facility includes two new lift pumps, each rated to provide flows of approximately 420 gpm, with space for a third pump. Both pumps operate at constant speed; however, they are equipped with variable frequency drives to allow operators to manually adjust the running speed to optimize pump cycle times. Typically, only one pump will operate at any given time, with a second pump for redundancy. However, during peak flows, both pumps are required, so it is recommended that the third pump be installed to provide redundancy during the peak flow events.

It is recommended that a minimum velocity of approximately 2.0 feet per second (fps) be maintained in the sewer force main to keep solids suspended so as not to accumulate at the bottom of the pipe. A peak flow velocity of at least 3.5 fps is desirable to re-suspend solids that have settled within the pipe. Table 5-2 presents sewer force main sizes and velocities at the lift station flows. The force main sizing will be re-evaluated once actual pump performance is known.

The new sewer force main will be 8-inch or 10-inch diameter, extending from the main sewer lift station to the headworks at the WWTP. The length of the proposed sewer force main is approximately 7,000 lineal feet. The force main will likely be constructed of C900 PVC pipe or fused HDPE pipe.

Table 5-2 Sewer Force Main Sizing

Lift Station Design Flows	Velocity	
	8-inch FM	10-inch FM
420 gpm (1 pump)	2.7 fps	1.8 fps
840 gpm (2 pumps)	5.4 fps	3.6 fps
1,260 gpm (3 pumps)	8.1 fps	5.4 fps

5.2.2 Headworks and Flow Metering Device

A new headworks structure will be constructed with the new process to accommodate the hydraulic requirements. A new automatically cleaned screen and a bypass channel with a manual bar screen will be installed. The existing headworks and pond will remain in service during the construction of the new facilities.

The existing ultrasonic flow meter should be replaced with the sewer force main replacement.

5.2.3 SCADA System

The existing WWTP does not have a Supervisory Control and Data Acquisition (SCADA) system. It is recommended that SCADA be considered with the WWTP upgrades to provide remote monitoring and alarm capabilities, as well as providing automatic reporting of critical information. The RPUD has already established a SCADA system, centralized at the District office, for its well sites and the new main sewer lift station. The WWTP can be integrated into the existing SCADA system if included in the project.

Installation of a SCADA system does not relieve the operator of the need to visit the WWTP on a daily basis. It would, however, provide the ability to view operations and received alarm indications when away from the WWTP, which could improve response time when problems arise.

In addition to SCADA at the WWTP, it is recommended that an alarm system be installed at the three lift stations within the sewer collection system. The alarm dialer system will notify the operators of high water levels and pump failures so these problems can be dealt with as soon as they occur.

5.2.4 Storage and Disposal Facilities

Existing disposal facilities include evaporation and percolation that occurs in the existing oxidation lagoons. Effluent from the oxidation lagoons is released to the adjacent reclamation area for disposal, as needed. The reclamation area is not currently harvested.

Effluent from the new treatment facilities will be discharged into a series of new storage and disposal ponds, as shown in Figure 5-1. Construction of the storage and disposal ponds will be phased, since 0.325 MGD of disposal capacity will not be necessary for several years. The first phase would include storage and disposal capacity for up to 0.275 MGD, which should be sufficient for more than 10 years, as shown on the wastewater flow projection chart in Figure 4-1. Some of the existing treatment ponds will be converted to storage and disposal ponds, as shown in Table 5-3, and the existing reclamation area will be replaced, since it is not currently being harvested.

Approximately 21.7 acres of new storage ponds will initially be constructed in order to meet an increased capacity to 0.275 MGD. As flows increase, future pond(s) will be constructed for discharge of the ultimate WWTP design capacity of 0.325 MGD flow. While disposal area for the entire treatment capacity is not anticipated to be necessary for at least 10 years, provisions will be constructed for these future facilities in the current project. The proposed storage ponds will be located to the west and north of the existing treatment lagoons. In addition, the existing Oxidation Lagoons 5 and 6 will be converted to storage and disposal ponds. Effluent from the proposed treatment facilities will flow by gravity into the proposed storage and disposal ponds.

Total storage volume of approximately 96 ac-ft and 24 acres (surface area) of disposal ponds will be provided for effluent disposal for 100-year return condition at effluent flows of 0.275 MGD as calculated in water balance spreadsheet (Appendix C). A summary of proposed storage pond sizes is shown in Table 5-3.

For the ultimate WWTP capacity of 0.325 MGD, an additional 4.6 acres of pond surface area will be required to supplement the evaporation and percolation from the initial phase ponds (see Appendix C). See Figure 5-1.

Use of recycled water for irrigation could alternatively be considered for disposal in the future. The secondary treatment level and the high EC levels in the effluent will limit the type of crops that could be used. Alfalfa irrigation would be suitable for use of secondary effluent for irrigation with the current effluent EC levels, although any increase in EC levels could impact the crop yield. Salt tolerance for alfalfa is approximately 2,000 μmhos/cm. If recycled water use is pursued, an agreement would need to be developed with a farmer to grow alfalfa, hay, or other forage or fodder crops for animals not producing milk for human consumption. If necessary, an effluent pump would be able to pump effluent from the ponds to the reclamation area. Use of recycled water for irrigation is not anticipated to be practicable at this time due to the low flows and relatively limited area that could be irrigated at those flows, and since there is no current agreement or demand for the use of recycled water. While percolation and evaporation ponds are recommended at this time, the proposed project will provide provisions for potential reclamation in the future. The potential of reclamation will be evaluated further in the Report of Waste Discharge.

Table 5-3 Proposed Storage and Disposal Ponds

Pond	Volume MG	Surface Area Acre	Water Depth ft
(E) Oxidation Lagoon 5 (Proposed Disposal Pond 5)	1.7	1.32	4
(E) Oxidation Lagoon 6 (Proposed Disposal Pond 6)	1.4	1.09	4
Proposed Pond 7	4.2	3.2	4
Proposed Pond 8	6.0	4.6	4
Proposed Pond 9	8.0	6.1	4
Proposed Pond 10	4.2	3.2	4
Proposed Pond 11	6.0	4.6	4
Total (Proposed)	**31.5**	**24.1**	
Future Pond 12	6.0	4.6	4
Total (Future)	**37.5**	**28.7**	

Prior to conversion of existing oxidation ponds to disposal ponds it is recommended that sludge be removed from each pond and ponds be disked to improve percolation.

A large volume of earthwork will be needed to construct the new storage and disposal ponds. It may be possible to raise the existing berm structure on the ponds to balance the cut and fill. However, the demand of soil for development may make soil export an economical option. Soil could be stored on the additional 33 acres owned by the District to the west of the WWTP.

5.2.5 Solids Handling

Solids handling facilities would not be required for the Aeration Pond alternative. However, for the other three alternatives considered, solids handling will be necessary. For the aeration pond alternative, sludge may be dredged and removed as necessary, likely after six to ten years of use, or more depending on influent loading. The removed sludge could be dried in a designated sludge drying area and hauled offsite. A sludge drying area would need to be concrete lined to prevent incidental percolation from the wet sludge. Alternatively, when sludge is removed, the contract dredging operations could include solids dewatering and hauling rather than drying and then hauling.

For the alternatives that include nitrification and denitrification, sludge needs to be wasted periodically from the treatment process. An aerobic sludge digester with fill-and-draw operation would be constructed. Positive displacement blowers would supply the aeration and mixing through coarse bubble diffusers. The digester would be sized to have 40 - 60 days of sludge detention time (depending on water temperature) to meet the requirements of Class B biosolids for disposal.

Supernatant from the sludge digester would be returned to the secondary process for further treatment while the thickened and stabilized sludge would be discharged to sludge drying beds that would be constructed.

The Class B biosolids requirements can also be met with the biosolids being air dried on paved drying beds for three months before hauling away for disposal. This could supplement the sludge digester to meet the Class B solids requirement when the sludge age is not met during certain weather conditions. Mechanical sludge dewatering could also be considered.

5.2.6 Electrical Facilities

The existing electrical facilities are old and likely insufficient for the proposed WWTP improvements. It is recommended that the electrical facilities be replaced with the proposed project.

5.2.7 Supply Water

The WWTP does not currently have supply water or any means of providing water for wash down of the facilities. Providing service from the District's potable water system would be cost prohibitive since it would require installation of about 1.5 miles of pipeline from the community to the WWTP.

It is recommended that a new well be installed at the WWTP to provide water for rinsing and wash down activities. Hose bibs would be provided near the headworks and treatment ponds for wash down of those facilities. A water supply will be necessary for rinsing of the proposed automatically cleaned screen at the headworks, and will be beneficial for other purposes too.

Riverdale has recently constructed new deep aquifer community supply wells for favorable groundwater that meet drinking water standards without the need for costly water treatment. The proposed well will likely be constructed to about 400 feet deep and may not meet drinking water standards. Since this will not be intended for drinking water purposes, appropriate signage will be included.

5.2.8 Building

A building with counter space and sink for sampling activities and a restroom may be considered. If the building is not included in this project, space will be designated for a building to be installed in the future.

5.2.9 Monitoring Wells

A groundwater monitoring well network may be required by the RWQCB. A groundwater monitoring well network would include installation of monitoring wells to monitor upgradient and downgradient shallow water quality in order to determine if the discharge is degrading groundwater quality. Three monitoring wells may be installed. A Groundwater Monitoring Well Installation Work Plan was submitted in December 2006. A letter from the RWQCB dated 14 February 2007 granted conditional approval of the Groundwater Monitoring Well Installation Work Plan which included a plan for installation of three monitoring wells.

Contract Documents for the monitoring wells were prepared and advertised in 2007, however no bids were received. Groundwater levels have declined since the Groundwater Monitoring Well Installation Work Plan was submitted in 2007. It is therefore recommended that the Work Plan be reviewed and updated as necessary, if monitoring wells are required.

The monitoring wells will be constructed along with the WWTP improvements, if required by the RWQCB.

5.2.10 Demolition of Abandoned Facilities

A clarifier digester (clarigester) and a trickling filter were constructed in 1958. These facilities have been abandoned in place since 1984. It is recommended that these facilities and associated piping and appurtenances be demolished as part of the proposed WWTP improvement project since they have not been used in over 30 years, and will not be used in the future.

The location of the existing clarigester and trickling filter is near the entrance of the facility, and would be a likely location of the operation building. In addition, the non-potable well is proposed to be constructed adjacent to these facilities.

5.2.11 Solar

Solar power is a renewable energy source that could save the RPUD on ongoing energy costs from the WWTP. Including solar would obviously increase the capital cost of the project, but it would reduce the energy consumption needs, and therefore provides the two main benefits (1) reducing energy costs for the RPUD, and (2) reducing the energy impacts of the new WWTP.

5.3 Treatment Process Alternatives

Four treatment processes have been evaluated, including: Aeration Pond System, Extended Aeration Activated Sludge Pond System (Biolac or Bioworks), STM Aerotor, and Membrane Bioreactor (MBR). The Aquamat treatment system was previously evaluated, but is no longer available as a wastewater treatment process in the United States, so it has been dropped from the current evaluation. The MBR system inherently provides higher-quality effluent than is required for this application, but capital costs for such systems have become more competitive and the potential increase in available disposal options warranted at least cursory review of that system.

It has been determined that nitrogen removal will not be required for the proposed WWTP improvements based on discussions with the RWQCB. Aeration pond systems provide secondary treatment without nitrogen removal. The other three alternatives evaluated herein have the capability of full nitrification and denitrification. Alternatives that include nitrification and denitrification are included because, before the preliminary antidegradation analysis was completed, it was not known whether or not it would be a requirement. Also, the cost comparison of alternatives is part of the justification for not providing nitrogen removal, from an economic standpoint. Due to the RWQCB's concern on the potential impact to groundwater quality from the infiltration of wastewater, all treatment ponds holding wastewater prior to the point of compliance will be lined. Downstream storage and disposal ponds will remain unlined.

5.3.1 Alternative 1 – Aeration Pond System

This alternative is a standard aeration pond system that consists of lined aeration ponds and disposal to evaporation/percolation ponds. The ponds would be equipped with surface aerators designed to provide aeration and complete mixing. This alternative does not include the use of clarifiers, so TSS will need to be controlled in the aeration ponds. The ponds will be designed to be intensely aerated and mixed, at over 20 horsepower (hp) per million gallons, which will eliminate the risk of temperature overturn or algae growth.

A quiescent zone can be designed to allow settling prior to discharge from the aeration ponds to the disposal ponds. This can be accomplished by the arrangement of the aerators or by installing curtains. The ponds may be designed with two outlets so the settling area can be switched periodically.

The process can be accomplished by constructing two new treatment ponds, each approximately 170 ft x 260 ft (2.4 MG). The new treatment ponds would each be equipped with five (5) 10 hp aerators. A third treatment pond would be constructed to provide redundancy. Aerators would only be installed in two ponds, and would be moved when the ponds are rotated.

The new treatment facilities can be constructed while the existing WWTP remains entirely in service.

Controls for the facilities are simple and based on a timer. The monitoring requirements will be similar to current operation. The addition of more surface aerators requires more mechanical maintenance.

5.3.2 Alternative 2 – Biolac or Bioworks Extended Aeration Activated Sludge System

This alternative is an extended aeration activated sludge system that consists of activated sludge facilities with nitrification/ denitrification, clarification, and disposal to evaporation/percolation ponds. The Biolac and Bioworks systems both incorporate lined treatment ponds and moving aeration headers and diffusers within the ponds. The aeration branch pipes float at the water surface and the aeration diffuser grids are suspended from the branch pipes. No components need to be installed on the basin bottom, therefore, the basin can potentially be retrofitted without dewatering or interruption of operation during construction. Positive-displacement blowers would be installed in a blower building to supply the air to the aeration manifold.

The mixed liquor from the aeration basin would flow to separate clarifiers by gravity. Biomass would be separated from the mixed liquor in the clarifiers. Settled biomass would be collected in the bottom of the clarifier and pumped by an air lift pump to the influent zone of the activated sludge basin. Biomass would be wasted periodically. Additional information regarding this treatment process is included in Appendix D.

The process can be accomplished in a single basin similar to what the District is presently operating, or in two parallel basins.

Controls for the facilities are relatively simple. The facilities could include dual clarifiers for redundancy and flexibility of operation.

5.3.3 Alternative 3 – STM Aerotor Activated Sludge System

The STM-Aerotor™ treatment process combines an Activated Sludge process and Fixed Film process in one tank. The fixed film media provides longer sludge age of the biomass for the nitrification process while a separate anoxic zone is provided for denitrification. Mounted on a center shaft, hollow media captures air as the shaft turns, drawing it down into the bottom of the tank. The captured air is slowly released as coarse bubbles. Oxygen transfer efficiency compares favorably to traditional coarse bubble aeration, but without the expense of running blowers or diffusers. Fixed film growth on the media is in an environment for nutrient removal, and would stay optimally reactive due to the roiling waters of the tank. Two submersible axial flow pumps recirculate the mixed liquor to the pre-anoxic tank to mix with the influent flow and achieve denitrification. Two mixers would be provided to meet the changing mixing need in the anoxic tank.

Two clarifiers following the STM aeration basin settle the solids and produce secondary effluent, which would be discharged to existing storage and disposal ponds. The chain and flight mechanism sweep the settled the sludge into a sludge sump inside the clarifier and a submersible return sludge pump returns sludge flow to the anoxic tank. Settled sludge would be periodically wasted to an aerobic digester, where the sludge would be digested and stabilized before drying. The supernatant from the sludge digester would be pumped back to the anoxic tank for further treatment.

The STM process offers the advantage of providing aeration by rotating fixed film paddles that allow reduction of the size of the blowers for the overall operation of the treatment facilities. Common wall construction of the anoxic basin, aeration basin, clarifier and digester reduces the structural cost of the concrete tanks. Dual clarifiers could be included for redundancy and flexibility of operation.

5.3.4 Alternative 4 – Membrane Bioreactor (MBR)

A Membrane Bioreactor process utilizes a membrane filter in lieu of the clarifiers in the traditional activated sludge plant. This allows for a smaller footprint, simplified separation process, and superior effluent quality, although the disposal methods for Riverdale WWTP do not necessarily need to achieve such a quality. A typical MBR plant consists of a fine screen headworks, equalization, anoxic basin, aeration basin and membrane filters. The membrane filter separates the biomass from the effluent and achieves an ultra-low-turbidity effluent equivalent to a traditional secondary process with filtration.

While many manufacturers have different layouts for the MBR package, one example of the package plant for the Riverdale WWTP consists of five container-sized, independently-operated small package plants, which in combination would provide for the ultimate 0.325 MGD flow. Each small plant contains all the necessary basins, equipment and controls needed to treat up to 75,000 GPD of sewage. The small plants can be installed incrementally with the increasing flow, making construction phasing very straightforward. The MBR plant requires a finer rotary screen than other alternatives to protect the membrane from physical damage from debris. The MBR plant also operates at much higher biomass concentration than the conventional processes, which further reduces the required tank volume and footprint. For process evaluation purpose, the cost comparison is based on full build-out capacity even though MBR process can be implemented in phases.

The MBR plant would also periodically waste sludge, although in slightly smaller quantities and relatively higher concentrations, to an aerobic digester for stabilization before dewatering. Additional blowers would be installed in a blower building to provide the aeration and mixing needed. Supernatant will be pumped to the flow distribution box to be further treated by the MBR process.

6 Selection of Alternative

Each of the alternatives considered has various advantages and disadvantages. This section will evaluate the costs and consider the strengths of each alternative to guide the recommendation of the most appropriate alternative for the Riverdale PUD WWTP.

6.1 Present Worth Cost Analysis

A present worth cost analysis is included as Table 6-1. The cost analysis presented in Table 6-1 is for comparison purposes only, and only includes items specific to the treatment processes. Additional costs will be included in the preliminary total cost estimate in Table 7-1. The present worth analysis also considers ongoing operation and maintenance costs. A more detailed breakdown of existing and projected O&M costs for the selected alternative is provided in Section 7.

Construction cost estimates are based on local bid canvasses and experience on local projects. The present worth calculations include a discount rate of 2 percent per year over 20 years of operations and maintenance costs. Present worth analysis indicates that the Aeration Pond system is the most cost-effective alternative.

Table 6-1 Present Worth Cost Analysis

		Treatment Processes				
Item No.	Item Description	Aeration Ponds	Biolac	Bioworks	STM Aerotor	MBR
General						
1	Mobilization, Bonds, Insurance	$220,000	$305,000	$305,000	$355,000	$505,000
2	Dust Control	$50,000	$50,000	$50,000	$50,000	$50,000
3	Worker Protection	$10,000	$10,000	$10,000	$10,000	$10,000
4	Clearing & Grubbing	$50,000	$50,000	$50,000	$50,000	$50,000
5	Demolition	$150,000	$150,000	$150,000	$150,000	$150,000
6	SWPPP Operations	$50,000	$50,000	$50,000	$50,000	$50,000
7	Miscellaneous Facilities & Operations	$30,000	$50,000	$50,000	$50,000	$50,000
Treatment System						
8	Force Main (on site) & Headworks	$186,000	$186,000	$186,000	$241,000	$266,000
9	Aeration Basin Structure	$902,000	$160,000	$135,000	$502,000	$133,000
10	Aeration Equipment	$200,000	$700,000	$310,500	$598,910	$4,427,000
11	Clarifier Structure	-	$257,500	$257,500	-	-
12	Clarifier Equipment	-	$369,200	$369,200	$287,585	-
13	RAS/WAS Pump Station	-	$80,000	$80,000	$30,000	-
14	Site and Yard Piping	$50,000	$137,800	$137,800	$77,600	$150,000
15	Aerobic Sludge Digester	-	$255,950	$255,950	$201,250	$224,000
16	Sludge Drying Beds	-	$300,000	$300,000	$300,000	$300,000

Item No.	Item Description		Aeration Ponds	Biolac	Bioworks	STM Aerotor	MBR
			Treatment Processes				
Site Work and Disposal Facilities							
17	Effluent Distribution Pipeline		$200,000	$200,000	$200,000	$200,000	$200,000
18	Storage and Disposal Facilities (Phase 1)		$580,000	$580,000	$580,000	$580,000	$580,000
19	Site Grading and Surfacing		$150,000	$150,000	$150,000	$150,000	$4,427,000
20	Fencing		$200,000	$200,000	$200,000	$200,000	-
21	Sludge Removal (existing ponds)		$250,000	$250,000	$250,000	$250,000	$250,000
22	Fill Existing Treatment Ponds		$95,000	$95,000	$95,000	$95,000	-
23	Monitoring Well Network Installation		$135,000	$135,000	$135,000	$135,000	$135,000
Electrical and Controls							
24	Electrical and Controls		$420,000	$660,700	$577,800	$989,200	$765,900
25	SCADA System Integration		$200,000	$200,000	$200,000	$200,000	$250,000
26	Emergency Power		$120,000	$120,000	$120,000	$120,000	$150,000
Subtotal for Secondary Effluent			4,248,000	5,704,000	5,207,000	5,650,000	9,141,000
	Engineering/Environmental/ Admin	24%	1,020,000	1,369,000	1,250,000	1,356,000	2,194,000
	Contingency	20%	850,000	1,141,000	1,041,000	1,130,000	1,828,000
Total For Secondary Treatment			**6,118,000**	**8,214,000**	**7,498,000**	**8,136,000**	**13,163,000**
	Annual O&M Cost		$345,000	445,500	413,500	479,800	880,600
	O&M Present Worth, 2%, 20 yr		5,641,000	7,285,000	6,761,000	7,845,000	14,399,000
	Total Present Worth		**11,759,000**	**15,499,000**	**14,259,000**	**15,981,000**	**27,562,000**

6.2 Planning Priorities

Part of the Government Code Section 65041.1 and sustainable water resource management priorities is to address the planning priorities. The following represent the priorities considered for this project.

6.2.1 Promoting Infill Development

This priority is to promote infill development and equity by rehabilitating, maintaining, and improving existing infrastructure that supports infill development and appropriate reuse and redevelopment of previously developed, underutilized land that is presently served by essential services, and to preserve cultural and historic resources.

This project will consist of utilizing an existing system. The land that this existing system is on will continue to be used to serve the Riverdale community, as it has been doing for over 50-years. This project will improve the existing sewage collection and treatment system and preserve the surrounding area for years to come.

6.2.2 Protect Environmental Resources

This priority is to protect environmental and agricultural resources by protecting groundwater quality.

The existing wastewater treatment system treats the wastewater to an unsatisfactory level in a series of complete-mix and oxidation ponds. The current effluent quality does not meet the existing Waste Discharge Requirements. The upgraded treatment plant will treat the BOD and TSS to a higher level, and improve the overall reliability of the discharge quality. Improvements made by this project would also minimize the risk of overflow of sewage due to aging infrastructure. This project is intended to protect one of the State's most valuable natural resources, being groundwater.

6.2.3 Encourage Efficient Development Patterns

As this project utilizes the existing system, this planning priority of encouraging efficient development patterns is met. In order to minimize costs to taxpayers, a project now that improves and rehabilitates the existing system will have a two-fold benefit.

First, the work performed now can minimize expenses in the future. As the work performed now will be utilizing the existing system, the system is not in such a detrimental state that it cannot be used. Waiting on this project could result in more expensive corrections in the future from a deteriorating wastewater system.

The second way costs to taxpayers can be minimized involves a preservation of groundwater quality by improving the effluent quality. Since the only source of potable water supply to the community is groundwater, the preservation of groundwater quality is critical to future potable supplies and the ability to avoid water treatment of the source water.

6.2.4 Encourage Sustainable Water Resources Management

This final priority encourages sustainable water resources management by ensuring that sustainable water resources measures, such as recycling wastewater, conserving water, conserving energy, and applying Low Impact Development Best Management Practices are utilized to the maximum extent possible.

The community based small size wastewater treatment plant minimizes the cost of wastewater collection and treatment to the public, yet provides efficient management.

6.3 Comparison of Alternatives

The alternatives described above are compared to determine the most appropriate alternative for the RPUD. The alternatives have been compared on the basis of life cycle costs, operation history, footprint, effluent quality, process stability, complexity, expandability, and operator familiarity. Each of these parameters is weighted based on the relative importance to the RPUD, and scored to determine the recommended alterative. A comparison matrix is presented in Table 6-2. The life cycle cost comparison is based on the present worth shown in Table 6-1, which takes into consideration the capital cost and ongoing operation and maintenance costs. Operation history takes into account the history of the treatment process and how long it has been in use, especially in the western United States. The footprint is the overall site area that the treatment system requires (higher score for a smaller area). Effluent quality is the quality of the treated effluent, especially with respect to BOD and TSS. Process stability is the ability to maintain steady and consistent treatment and effluent quality. Complexity refers to how complex the treatment system is to operate. Expandability is the ability to construct in a modular design to accommodate future expansion. Operator familiarity is how familiar the operator is with the process, and how it compares with the operation of the existing WWTP.

None of the alternatives evaluated require complete bypass of the existing treatment process during construction. The existing facilities are required to be in continuous service in order to maintain compliance with Waste Discharge Requirements throughout construction.

Alternative 1 (Aeration Pond System) would be constructed north of the existing treatment ponds, and would not impact the existing system during construction. This alternative would offer operational stability to shock loading with the long detention time. It would utilize intense aeration and complete mixing capabilities to eliminate the risk of temperature overturn or algae growth. A quiescent zone would be incorporated to allow for settling prior to discharge. This alternative does not include sludge removal or handling facilities. However, sludge would likely only need to be removed after 10 to 20 years of operation. Also with three treatment ponds, one pond can be removed from service for sludge removal or liner maintenance, without impacting the treatment performance. Dredging can also be done without interrupting the operation of the ponds. This alternative has the lowest life cycle cost, and has significant operation history. It does, however, require the largest footprint of all the alternatives.

Alternative 2a (Biolac) would be constructed adjacent and in addition to the existing treatment process and would not impact the existing system during construction. This alternative would offer operational stability in that it would utilize a long sludge age. Biological nutrient reduction would be accomplished with control of air flow distribution in the basin with multiple oxic and anoxic zones. Separate circular clarifiers would be located at one end of the basin. Sludge would be further treated in an aerobic digester and air-dried in sludge drying beds. Dried solids would be stored onsite in a concrete-lined storage bin before hauling away for disposal. This alternative provides good effluent quality and process stability, and is less complex than the STM and MBR alternatives, but it is more expensive than the Aeration Pond system.

Alternative 2b (Bioworks) would be constructed adjacent to the existing CM lagoon, and would not impact the existing system during construction. This alternative would also offer operational stability in that it would also utilize a long sludge age. Biological nutrient reduction would be accomplished in a similar manner to the Biolac process as well, with control of air flow distribution in the basin using multiple oxic and anoxic zones. One additional feature Bioworks offers is the ability to control a variable-buoyancy floating chamber in each grid to allow raising the diffuser grid for maintenance. Separate circular clarifiers would be located at one end of the basin. This alternative is very similar to the Biolac, except it can be accomplished in a smaller footprint. The Bioworks has not been around as long as Biolac, but utilizes a very similar technology.

Alternative 3 (STM-Aerotor) would be constructed in a rectangular reinforced-concrete structure adjacent to the existing treatment facility, and would not impact the existing system during construction. This alternative would offer operational flexibility because it consists of multiple aeration and clarifier tanks. Nitrification and denitrification take place in a separate pre-anoxic basin. This alternative would also include an aerobic sludge digestion chamber within the same structure. Sludge drying beds and a biosolids storage bin would be constructed. This alternative provides good effluent quality, but it is less stable, more complex, and has a higher life cycle cost than the Aeration Pond and Biolac/Bioworks alternatives.

Alternative 4 (MBR) would consist of five containerized treatment packages installed on a concrete pad adjacent to the existing treatment facility, and would not impact the existing system during construction. This alternative would offer the flexibility of phasing the construction to purchase three or four containers (up to 0.225 or 0.30 mgd, respectively) in the beginning and install additional treatment packages as needed. There would be a headworks structure with fine screening and a distribution box to distribute the flow to each treatment package, and to allow for segregation of any treatment package for maintenance purposes. The treatment process is self-contained requiring only influent, effluent and sludge connections, and power supply. A separate aerobic sludge digester and sludge drying beds would be constructed, similar to Alternatives 1a and 1 b. This alternative is significantly more complex and has a higher life cycle cost than the other alternatives. While it would produce the highest quality effluent, effluent treated to the level that the MBR produces is not necessary for RPUD.

Preliminary process layouts for each of the alternatives are included as Figure 6-1 through Figure 6-5.

Table 6-2 Alternatives Comparison Matrix

Parameter	Weight	Aeration Ponds	Biolac	Bioworks	STM	MBR
Life cycle cost	30%	10	7	8	6	4
Ops History	15%	10	8	7	7	7
Footprint	5%	5	7	10	8	9
Effluent quality	10%	7	9	9	9	10
Process stability	10%	10	10	10	7	8
Complexity	10%	10	8	8	7	5
Expandability	10%	7	8	9	7	10
Operator Familiarity	10%	10	9	9	8	7
Total	100%	9.2	8.1	8.5	7.1	6.7

7 Proposed Project (Recommended Alternative)

7.1 Project Description

Based on the comparison matrix, Alternative 1, Aeration Pond System is the preferred process for the WWTP upgrade. The Aeration Pond System would include construction of three new treatment ponds north of the existing system (see Figure 6-1). A new headworks would be constructed with an automatically cleaned screen. Other ancillary facilities would include a new flow meter and MCC.

A preliminary flow diagram and preliminary hydraulic profile are included in Figure 7-1 and Figure 7-2.

7.2 Operations During Construction

The WWTP improvements will be constructed without interrupting the existing plant operation. There will be only a short system downtime when the flow is transferred from the existing to the new treatment system. This downtime would be coordinated with the RPUD and could be handled by backing up the lift station for a short period of time.

7.3 Total Project Cost Estimate

The preliminary project cost estimate for the proposed WWTP improvements that have been discussed is shown as Table 7-1. Project cost estimates will be updated and refined during design of the facilities. As costs are refined, some of the additional items listed may become additive alternates that may depend on actual bid costs and the amount and type of funding available.

Table 7-1 Preliminary Total Project Cost Estimate

Item No.	Item Description		
General			
1	Mobilization, Bonds, Insurance		$220,000
2	Dust Control		$50,000
3	Worker Protection		$10,000
4	Clearing & Grubbing		$50,000
5	Demolition		$150,000
6	SWPPP Operations		$50,000
7	Miscellaneous Facilities and Operations		$30,000
Treatment System			
8	Force Main (on site) & Headworks		$186,000
9	Aeration Basin Structure		$902,000
10	Aeration Equipment		$200,000
11	Clarifier Structure		$0
12	Clarifier Equipment		$0
13	RAS/WAS Pump Station		$0
14	Site and Yard Piping		$50,000
15	Aerobic Sludge Digester		$0
16	Sludge Drying Beds		$0
Site Work and Disposal Facilities			
17	Effluent Distribution Pipeline		$200,000
18	Storage and Disposal Facilities (Phase 1)		$580,000
19	Site Grading and Surfacing		$150,000
20	Fencing		$200,000
21	Sludge Removal (existing ponds)		$250,000
22	Fill Existing Treatment Ponds		$95,000
23	Monitoring Well Network Installation		$135,000
Electrical and Controls			
24	Electrical and Controls		$420,000
25	SCADA System Integration		$200,000
26	Emergency Power		$120,000
Construction Subtotal - Base Items			$4,248,000
	Contingency	20%	$850,000
Total Estimated Construction Cost - Base Items			**$5,098,000**

Item No.	Item Description		
Additional Items			
27	Sewer Force Main Replacement		$590,000
28	Supply Well Installation and Equipping		$150,000
29	Supply Water Site Piping		$70,000
30	Access Road Improvements		$250,000
31	Lift Station Pump and VFD		$100,000
32	Lift Station Alarm Dialer Systems (Optional)		$250,000
33	Sludge Drying Beds (Optional)		$300,000
34	Operations Building (Optional)		$200,000
35	Solar (Optional)		$750,000
Construction Subtotal			$2,650,000
	Contingency	20%	$530,000
Total Estimated Construction Cost			**$8,278,000**
	Engineering/Permitting[1]	10%	$828,000
	Environmental		$60,000
	Admin/Legal		$80,000
	PG&E Rule 16		$60,000
	Survey		$75,000
	Bidding and Advertisement		$45,000
	Labor Compliance		$30,000
	Construction Services		$425,000
Non-Construction Subtotal			$1,603,000
Non-Construction Subtotal less $500k from Planning Funding			$1,103,000
Total Estimated Construction Phase Project Cost			**$9,381,000**

1. A portion of the engineering, permitting and environmental costs shown will be covered under the planning funding ($500,000).

7.4 Annual Operating Budget

The proposed WWTP improvements will result in increased annual operating costs, including increase in power costs, equipment maintenance, and equipment replacement. Table 7-2 presents a summary of estimated operating costs.

If solar is included in the project, the power costs will be reduced. However, there would be an increase in equipment maintenance and replacement, and reserve set aside. Table 7-3 presents a summary of estimated operating costs if solar panels are installed. The estimated power costs are based on 80% of the projected demand produced by the solar panels, with a performance ratio of 79%.

Table 7-2 Preliminary Estimate of Annual Operating Costs

Cost Component		
Power Cost (WWTP)	0.12/kW-hr	$ 84,500
Equipment Maintenance		20,000
Chemical Cost		0
Labor		75,000
Insurance		12,500
Dues Subscriptions		18,000
Reserves for Replacement*		55,000
Office Staff		25,000
Supplies		15,000
Legal and Professional Fees		40,000
Total Annual Cost		**$ 345,000**

*Note: The operating expenses shown do not include Depreciation, but there is a line item for Reserves for Replacement of equipment.

Table 7-3 Preliminary Estimate of Annual Operating Costs with Solar

Cost Component	
Power Cost (WWTP)	$ 31,000
Equipment Maintenance	25,500
Chemical Cost	0
Labor	75,000
Insurance	12,500
Dues Subscriptions	18,000
Reserves for Replacement	70,000
Office Staff	25,000
Supplies	15,000
Legal and Professional Fees	40,000
Total Annual Cost	**$ 312,000**

The current annual operating revenues from the District's sewer fund are about $475,793, based on the year ended June 30, 2015 financial report. The operating expenses are about $288,580 (see Table 7-4). The proposed treatment alternatives would cause a minimal increase in annual operations and maintenance expenses, including increased power, equipment maintenance and replacement, and labor costs. Based on the current revenue from the sewer fund and the preliminary estimate of annual operating costs, it is anticipated that the proposed improvements could be implemented without need for a rate increase. The operating cost estimate should be refined during design to determine whether a rate study is necessary.

The RPUD plans to apply for CWSRF construction funding for the proposed improvements. Since the RPUD is severely disadvantaged, based on the median household income (MHI) being less than 60% of the statewide average, and it is considered a small community (population less than 20,000), it is anticipated that they would be eligible for grant funding. The amount of grant funding and terms will be determined during the review of a construction funding application.

RPUD's current sewer rates are $39 per month ($468 per year). The current MHI is $30,159, based on an income survey conducted in 2013. The sewer rates are therefore currently approximately 1.6% of the MHI. The sewer rates were raised to $39 per month for wastewater and sewer improvements. The RPUD used those funds for a needed sewer lift station project. The sewer lift station project was constructed primarily in 2016, so the use of those funds is not reflected in the 2015 financial statement below.

Table 7-4 Sewer Fund 2015 Financial Statement

	Sewer Fund
Operating Revenues	
Service Revenues	$475,793
Penalties	$0
Total Revenues	$475,793
Operating Expenses	
Payroll and Benefits	$90,075
Depreciation	$28,537
Dues Subscriptions	$17,909
Insurance	$14,914
Legal and Professional Fees	$61,240
Repairs and Maintenance	$10,974
Services-Other	$0
Supplies	$12,367
Utilities	$52,564
Total Expenditures	$288,580
Operating Income (Loss)	$187,213
Non-operating Revenues (Expenses)	
Investment Income (Loss)	$0
Miscellaneous Income	$0
Tax Revenues	$11,100
Interest Expense	($3,355)
Total Non-operating Revenues (Expenses)	$7,745
Income (Loss) Before Capital Contributions	$194,958
Capital Contributions	$0
Changes in Net Position	$194,958
Net Position at Beginning of Year	$1,876,173
Net Position at End of Year	$2,071,131

Source: Riverdale Public Utility District Financial Statements with Accompanying Information and Independent Auditor's Report for the Year Ended June 30, 2015.

7.5 Additional Considerations

7.5.1 Environmental Issues

Environmental compliance documents will be prepared for the proposed project. Environmental documents will be prepared for compliance with the California Environmental Quality Act (CEQA) and federal crosscutting requirements to comply with funding program requirements that include federal funds. It is anticipated that an Initial Study/Mitigated Negative Declaration will be the appropriate level of environmental document required for this project.

This will include preparation of the documents, issuing public notice, circulating the Initial Study/Mitigated Negative Declaration for public comment, and holding a public hearing prior to adopting the Initial Study/Mitigated Negative Declaration.

7.5.2 Permits and Agreements

A Report of Waste Discharge has been prepared for the proposed RPUD WWTP improvements, and submitted to the RWQCB for review and approval, and issuance of new Waste Discharge Requirements. An Antidegradation Analysis has been prepared and is appended to the Report of Waste Discharge submitted to the RWQCB.

The WWTP improvement project may also involve other permits, including an Encroachment Permit, Grading Permit, Conditional Use Permit, or others.

If reclamation of wastewater effluent is considered in the future, a Title 22 Engineering Report (Report of Water Reclamation) would need to be prepared and submitted to the Division of Drinking Water. This is not proposed with the proposed project.

7.5.3 Utility Services

The electrical service will need to be upgraded to 480V/3phase.

Water for wash down will be needed; however extending a water service from the Riverdale PUD would be cost prohibitive due to the distance from the District. It is instead recommended that a water supply well be drilled at the WWTP site for wash down purposes.

7.5.4 Right-of-Way or Easement Requirements

The proposed sewer force main replacement will be constructed within an existing 20 foot easement between the lift station and the WWTP.

7.5.5 Operator Requirements

The WWTP improvements will require a Grade 2 operator. The Riverdale PUD operator currently has a Grade 1 certification. A Grade 2 operator certification will be pursued.

7.6 Schedule

The Report of Waste Discharge, Antidegradation Analysis, and Environmental Documents are planned to be completed by February 2017. Construction Documents prepared to a 90 percent level will be completed in 2017. The District will apply for and secure funding to construct the proposed WWTP improvement project before Construction Documents can be finalized to advertise, bid, and construct the project.

Appendix

Appendix A
CIWQS Violations Report

Facility At-A-Glance Report

SEARCH CRITERIA:

Place ID **273185**

General Information

Region	Place ID	Place Name	Place Type	Place Address	Place County
5F	273185	Riverdale WWTF	Wastewater Treatment Facility	20896 Malsbary Riverdale, CA, 93656-9274	Fresno

Related Parties

Party	Party Type	Party Name	Role	Classification	Relationship Start Date	Relationship End Date
81799	Person	Ron Bass	Contact		06/17/2005	
37493	Organization	Riverdale PUD	Owner	Special District	03/24/1958	

Total Related Parties: 2

Regulatory Measures

Reg Measure ID	Reg Measure Type	Region	Program	Order No.	WDID	Effective Date	Expiration Date	Status	Amended?
144572	WDR	5F	WDRMUNILRG	85-252	5D100114001	09/27/1985	09/27/1995	Active	N
143567	WDR	5F	WDRMUNILRG	58-023	5D100114001	03/24/1958	03/24/1963	Historical	N

Total Reg Measures: 2

Violations

Violation ID	Occurred Date	Violation Type	(-) Violation Description	Corrective Action	Status	Classification	Source
1007782	04/30/2016	DMON	April 2016 SMR missing daily flow (2x).		Violation	3	Report
1007754	04/30/2016	Order Conditions	4M; EC; 1100; umhos/cm; IM; 1220 - 1390 (4x)		Violation	3	Report
1007657	04/30/2016	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 180.0 mg/L.		Violation	3	Report
1007680	04/30/2016	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 180.0 mg/L.		Violation	3	Report
1007781	03/31/2016	DMON	March 2016 SMR missing daily flow (31x).		Violation	3	Report
1007753	03/31/2016	Order Conditions	3M; EC; 1100; umhos/cm; IM; 1123 - 1303 (5x)		Violation	3	Report
1007656	03/31/2016	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 65.0 mg/L.		Violation	3	Report
1007780	02/29/2016	DMON	February 2016 SMR missing daily flow (29x).		Violation	3	Report
1007752	02/29/2016	Order Conditions	2M; EC; 1100; umhos/cm; IM; 1166 - 1288 (2x)		Violation	3	Report
1007655	02/29/2016	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 62.0 mg/L.		Violation	3	Report
1007778	01/31/2016	DMON	January 2016 SMR missing daily flow (2x).		Violation	3	Report
1007751	01/31/2016	Order Conditions	1M; EC; 1100; umhos/cm; IM; 1139 (1x)		Violation	3	Report
1007654	01/31/2016	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 180.0 mg/L.		Violation	3	Report
1007679	01/31/2016	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 180.0 mg/L.		Violation	3	Report

1007750	12/31/2015	Order Conditions	12M; EC; 1100; umhos/cm; IM; 1145 - 1250 (4x)	Violation	3	Report
1007653	12/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 180.0 mg/L.	Violation	3	Report
1007678	12/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 180.0 mg/L.	Violation	3	Report
1007749	11/30/2015	Order Conditions	11M; EC; 1100; umhos/cm; IM; 1225 - 1330 (4x)	Violation	3	Report
1007652	11/30/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 93.0 mg/L.	Violation	3	Report
1007677	11/30/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 93.0 mg/L.	Violation	3	Report
1007748	10/31/2015	Order Conditions	10M; EC; 1100; umhos/cm; IM; 1209 - 1698 (4x)	Violation	3	Report
1007651	10/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 320.0 mg/L.	Violation	3	Report
1007676	10/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 320.0 mg/L.	Violation	3	Report
1007704	08/31/2015	Order Conditions	8M; DO; >1.0; mg/L; IM; 0.3 - 0.9 (6x)	Violation	3	Report
1007747	08/31/2015	Order Conditions	8M; EC; 1100; umhos/cm; IM; 1735 - 1819 (4x)	Violation	3	Report
1007650	08/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 82.0 mg/L.	Violation	3	Report
1007675	08/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 82.0 mg/L.	Violation	3	Report
1007776	07/31/2015	DMON	June 2015 SMR missing daily flow (30x).	Violation	3	Report
1007703	07/31/2015	Order Conditions	7M; DO; >1.0; mg/L; IM; 0.9 (1x)	Violation	3	Report
1007746	07/31/2015	Order Conditions	7M; EC; 1100; umhos/cm; IM; 1756 - 2010 (5x)	Violation	3	Report
1007649	07/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 56.0 mg/L.	Violation	3	Report
1007775	06/30/2015	DMON	June 2015 SMR missing daily flow (30x).	Violation	3	Report
1007701	06/30/2015	Order Conditions	6M; DO; >1.0; mg/L; IM; 0.6 (1x)	Violation	3	Report
1007745	06/30/2015	Order Conditions	6M; EC; 1100; umhos/cm; IM; 1583 - 1772 (4x)	Violation	3	Report
1007648	06/30/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 120.0 mg/L.	Violation	3	Report
1007674	06/30/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 120.0 mg/L.	Violation	3	Report
1007744	05/31/2015	Order Conditions	5M; EC; 1100; umhos/cm; IM; 1400 - 1516 (4x)	Violation	3	Report
1007647	05/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 100.0 mg/L.	Violation	3	Report
1007673	05/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 100.0 mg/L.	Violation	3	Report
1007774	04/30/2015	DMON	April 2015 SMR missing daily flow (30x).	Violation	3	Report
1007743	04/30/2015	Order Conditions	4M; EC; 1100; umhos/cm; IM; 1275 - 1511 (5x)	Violation	3	Report
1007646	04/30/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 100.0 mg/L.	Violation	3	Report
1007672	04/30/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and	Violation	3	Report

			reported value was 100.0 mg/L.			
1007700	03/31/2015	Order Conditions	3M; DO; >1.0; mg/L; IM; 0.3 - 0.6 (2x)	Violation	3	Report
1007742	03/31/2015	Order Conditions	3M; EC; 1100; umhos/cm; IM; 1191 - 1333 (4x)	Violation	3	Report
1007645	03/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 130.0 mg/L.	Violation	3	Report
1007671	03/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 130.0 mg/L.	Violation	3	Report
1007772	02/28/2015	DMON	February 2015 SMR missing daily flow (28x).	Violation	3	Report
1007699	02/28/2015	Order Conditions	2M; DO; >1.0; mg/L; IM; 0.3 - 0.8 (3x)	Violation	3	Report
1007741	02/28/2015	Order Conditions	2M; EC; 1100; umhos/cm; IM; 1190 - 1268 (3x)	Violation	3	Report
1007770	01/31/2015	DMON	January 2015 SMR missing weekly DO (1x), weekly EC (1x).	Violation	3	Report
1007698	01/31/2015	Order Conditions	1M; DO; >1.0; mg/L; IM; 0.6 (1x)	Violation	3	Report
1007740	01/31/2015	Order Conditions	1M; EC; 1100; umhos/cm; IM; 1116 - 1293 (3x)	Violation	3	Report
1007644	01/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 98.0 mg/L.	Violation	3	Report
1007670	01/31/2015	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 98.0 mg/L.	Violation	3	Report
1007739	12/31/2014	Order Conditions	12M; EC; 1100; umhos/cm; IM; 1117 - 1227 (4x)	Violation	3	Report
1007643	12/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 87.0 mg/L.	Violation	3	Report
1007669	12/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 87.0 mg/L.	Violation	3	Report
1007697	11/30/2014	Order Conditions	11M; DO; >1.0; mg/L; IM; 0.8 (1x)	Violation	3	Report
1007736	11/30/2014	Order Conditions	11M; EC; 1100; umhos/cm; IM; 1211 - 1332 (4x)	Violation	3	Report
1007642	11/30/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 69.0 mg/L.	Violation	3	Report
1007768	10/31/2014	DMON	October 2014 SMR missing weekly DO (1x), weekly EC (1x).	Violation	3	Report
1007696	10/31/2014	Order Conditions	10M; DO; >1.0; mg/L; IM; 0.8 - 0.9 (2x)	Violation	3	Report
1007718	10/31/2014	Order Conditions	10M; EC; 1100; umhos/cm; IM; 1324 - 1476 (4x)	Violation	3	Report
1007641	10/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 140.0 mg/L.	Violation	3	Report
1007668	10/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 140.0 mg/L.	Violation	3	Report
1007695	09/30/2014	Order Conditions	9M; DO; >1.0; mg/L; IM; 0.3 - 0.9 (11x)	Violation	3	Report
1007717	09/30/2014	Order Conditions	9M; EC; 1100; umhos/cm; IM; 1390 - 1593 (4x)	Violation	3	Report
1007640	09/30/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 76.0 mg/L.	Violation	3	Report
1007694	08/31/2014	Order Conditions	8M; DO; >1.0; mg/L; IM; 0.3 - 0.8 (10x)	Violation	3	Report
1007716	08/31/2014	Order Conditions	8M; EC; 1100; umhos/cm; IM; 1121 - 1417 (4x)	Violation	3	Report
1007639	08/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 96.0 mg/L.	Violation	3	Report

1007693	07/31/2014	Order Conditions	7M; DO; >1.0; mg/L; IM; 0.6 - 0.9 (9x)	Violation	3	Report
1007715	07/31/2014	Order Conditions	7M; EC; 1100; umhos/cm; IM; 1475 - 1561 (2x)	Violation	3	Report
1007638	07/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 58.0 mg/L.	Violation	3	Report
1007767	06/30/2014	DMON	June 2014 SMR missing weekly EC (4x).	Violation	3	Report
1007637	06/30/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 61.0 mg/L.	Violation	3	Report
1007766	05/31/2014	DMON	May 2014 SMR missing weekly EC (4x).	Violation	3	Report
1007636	05/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 45.0 mg/L.	Violation	3	Report
1007714	04/30/2014	Order Conditions	4M; EC; 1100; umhos/cm; IM; 1159 - 1249 (4x)	Violation	3	Report
1007635	04/30/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 94.0 mg/L.	Violation	3	Report
1007667	04/30/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 94.0 mg/L.	Violation	3	Report
1007765	03/31/2014	DMON	March 2014 SMR missing weekly EC (4x).	Violation	3	Report
1007692	03/31/2014	Order Conditions	3M; DO; >1.0; mg/L; IM; 0.4 - 0.8 (5x)	Violation	3	Report
1007634	03/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 100.0 mg/L.	Violation	3	Report
1007666	03/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 100.0 mg/L.	Violation	3	Report
1007691	02/28/2014	Order Conditions	2M; DO; >1.0; mg/L; IM; 0.3 - 0.8 (2x)	Violation	3	Report
1007633	02/28/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 74.0 mg/L.	Violation	3	Report
1007764	01/31/2014	DMON	January 2014 SMR missing weekly EC (5x).	Violation	3	Report
1007690	01/31/2014	Order Conditions	1M; DO; >1.0; mg/L; IM; 0.3 - 0.9 (5x)	Violation	3	Report
1007632	01/31/2014	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 71.3 mg/L.	Violation	3	Report
1007689	12/31/2013	Order Conditions	12M; DO; >1.0; mg/L; IM; 0.2 - 0.7 (2x)	Violation	3	Report
1007631	12/31/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 120.0 mg/L.	Violation	3	Report
1007664	12/31/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 120.0 mg/L.	Violation	3	Report
1007630	11/30/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 54.0 mg/L.	Violation	3	Report
1007763	10/31/2013	DMON	October 2013 SMR missing weekly EC (1x), daily flow (2x).	Violation	3	Report
1007688	10/31/2013	Order Conditions	10M; DO; >1.0; mg/L; IM; 0.8 - 0.9 (3x)	Violation	3	Report
1007629	10/31/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 54.0 mg/L.	Violation	3	Report
1007687	09/30/2013	Order Conditions	9M; DO; >1.0; mg/L; IM; 0.4 - 0.9 (6x)	Violation	3	Report
1007713	09/30/2013	Order Conditions	9M; EC; 1100; umhos/cm; IM; 1153 - 1329 (4x)	Violation	3	Report
1007628	09/30/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 48.0 mg/L.	Violation	3	Report
			August 2013 SMR missing weekly DO (1x),			

1007762	08/31/2013	DMON	weekly EC (1x).	Violation	3	Report
1007712	08/31/2013	Order Conditions	8M; EC; 1100; umhos/cm; IM; 1321 - 1442 (3x)	Violation	3	Report
1007627	08/31/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 44.0 mg/L.	Violation	3	Report
1007686	07/31/2013	Order Conditions	7M; DO; >1.0; mg/L; IM; 0.1 - 0.8 (11x)	Violation	3	Report
1007711	07/31/2013	Order Conditions	7M; EC; 1100; umhos/cm; IM; 1263 - 1471 (5x)	Violation	3	Report
1007685	06/30/2013	Order Conditions	6M; DO; >1.0; mg/L; IM; 0.4 - 0.9 (2x)	Violation	3	Report
1007710	06/30/2013	Order Conditions	6M; EC; 1100; umhos/cm; IM; 1269 - 1564 (4x)	Violation	3	Report
1007709	05/31/2013	Order Conditions	5M; EC; 1100; umhos/cm; IM; 1109 (1x)	Violation	3	Report
1007626	05/31/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 56.0 mg/L.	Violation	3	Report
1007761	04/30/2013	DMON	April 2013 SMR missing daily flow (5x).	Violation	3	Report
1007625	04/30/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 120.0 mg/L.	Violation	3	Report
1007663	04/30/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 120.0 mg/L.	Violation	3	Report
1007624	03/31/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 110.0 mg/L.	Violation	3	Report
1007662	03/31/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 110.0 mg/L.	Violation	3	Report
1007623	01/31/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 97.0 mg/L.	Violation	3	Report
1007661	01/31/2013	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 97.0 mg/L.	Violation	3	Report
1007622	12/31/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 110.0 mg/L.	Violation	3	Report
1007660	12/31/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 110.0 mg/L.	Violation	3	Report
1007760	11/30/2012	DMON	November 2012 SMR missing weekly DO (1x), weekly EC (1x).	Violation	3	Report
1007758	11/30/2012	OEV	Flow 30-Day Average limit is 0.250 million gallons and reported value was 0.256 million gallons.	Violation	3	Report
1007621	11/30/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 92.0 mg/L.	Violation	3	Report
1007659	11/30/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 92.0 mg/L.	Violation	3	Report
1007684	10/31/2012	Order Conditions	10M; DO; >1.0; mg/L; IM; 0.4 - 0.9 (7x)	Violation	3	Report
1007708	10/31/2012	Order Conditions	10M; EC; 1100; umhos/cm; IM; 1108 - 1193 (4x)	Violation	3	Report
1007759	10/31/2012	OEV	Flow 30-Day Average limit is 0.250 million gallons and reported value was 0.254 million gallons.	Violation	3	Report
1007620	10/31/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 89.0 mg/L.	Violation	3	Report
1007658	10/31/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Maximum limit is 80.0 mg/L and reported value was 89.0 mg/L.	Violation	3	Report
1007683	09/30/2012	Order Conditions	9M; DO; >1.0; mg/L; IM; 0.4 - 0.9 (12x)	Violation	3	Report
1007707	09/30/2012	Order	9M; EC; 1100; umhos/cm; IM; 1190 - 1235 (3x)	Violation	3	Report

		Conditions				
1007757	09/30/2012	OEV	Flow 30-Day Average limit is 0.250 million gallons and reported value was 0.254 million gallons.	Violation	3	Report
1007706	08/31/2012	Order Conditions	7M; EC; 1100; umhos/cm; IM; 1240 - 1293 (5x)	Violation	3	Report
1007682	08/31/2012	Order Conditions	8M; DO; >1.0; mg/L; IM; 0.3 - 0.9 (17x)	Violation	3	Report
1007756	08/31/2012	OEV	Flow 30-Day Average limit is 0.250 million gallons and reported value was 0.252 million gallons.	Violation	3	Report
1007681	07/31/2012	Order Conditions	7M; DO; >1.0; mg/L; IM; 0.5 - 0.9 (10x)	Violation	3	Report
1007705	07/31/2012	Order Conditions	7M; EC; 1100; umhos/cm; IM; 1226 - 1263 (4x)	Violation	3	Report
1007619	07/31/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average (Mean) limit is 40.0 mg/L and reported value was 41.0 mg/L.	Violation	3	Report
931127	06/30/2012	OEV	Dissolved Oxygen Instantaneous Minimum limit is 1 ml/L and reported value was .6 ml/L.	Violation	3	Report
931058	06/30/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average limit is 40 mg/L and reported value was 50 mg/L.	Violation	3	Report
931077	06/22/2012	OEV	Electrical Conductivity @ 25 Deg. C Instantaneous Maximum limit is 1100 umhos/cm and reported value was 1203 umhos/cm.	Violation	3	Report
931126	05/31/2012	OEV	Dissolved Oxygen Instantaneous Minimum limit is 1 ml/L and reported value was .2 ml/L.	Violation	3	Report
931076	05/16/2012	OEV	Electrical Conductivity @ 25 Deg. C Instantaneous Maximum limit is 1100 umhos/cm and reported value was 1172 umhos/cm.	Violation	3	Report
931125	04/30/2012	OEV	Dissolved Oxygen Instantaneous Minimum limit is 1 ml/L and reported value was .6 ml/L.	Violation	3	Report
931057	04/30/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average limit is 40 mg/L and reported value was 81 mg/L.	Violation	3	Report
931056	04/02/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Daily Maximum limit is 80 mg/L and reported value was 81 mg/L.	Violation	3	Report
931124	03/31/2012	OEV	Dissolved Oxygen Instantaneous Minimum limit is 1 ml/L and reported value was .5 ml/L.	Violation	3	Report
931135	01/31/2012	Deficient Reporting	January 2012 SMR missing weekly DO x1.	Violation	3	Report
931123	01/31/2012	OEV	Dissolved Oxygen Instantaneous Minimum limit is 1 ml/L and reported value was .3 ml/L.	Violation	3	Report
931055	01/31/2012	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average limit is 40 mg/L and reported value was 51 mg/L.	Violation	3	Report
931053	12/31/2011	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average limit is 40 mg/L and reported value was 92 mg/L.	Violation	3	Report
931054	12/05/2011	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Daily Maximum limit is 80 mg/L and reported value was 92 mg/L.	Violation	3	Report
931052	11/30/2011	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average limit is 40 mg/L and reported value was 54 mg/L.	Violation	3	Report
931075	10/26/2011	OEV	Electrical Conductivity @ 25 Deg. C Instantaneous Maximum limit is 1100 umhos/cm and reported value was 1190 umhos/cm.	Violation	3	Report
931074	09/21/2011	OEV	Electrical Conductivity @ 25 Deg. C Instantaneous Maximum limit is 1100 umhos/cm and reported value was 1256 umhos/cm.	Violation	3	Report
931122	09/14/2011	OEV	Dissolved Oxygen Instantaneous Minimum limit is 1 ml/L and reported value was .7 ml/L.	Violation	3	Report
931121	08/31/2011	OEV	Dissolved Oxygen Instantaneous Minimum limit is 1 ml/L and reported value was .8 ml/L.	Violation	3	Report
931073	08/10/2011	OEV	Electrical Conductivity @ 25 Deg. C Instantaneous Maximum limit is 1100 umhos/cm and reported value was 1319 umhos/cm.	Violation	3	Report
			Electrical Conductivity @ 25 Deg. C			

931072	07/06/2011	OEV	Instantaneous Maximum limit is 1100 umhos/cm and reported value was 1348 umhos/cm.	Violation	3	Report
931051	06/30/2011	CAT1	Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) Monthly Average limit is 40 mg/L and reported value was 75 mg/L.	Violation	3	Report
931071	06/22/2011	OEV	Electrical Conductivity @ 25 Deg. C Instantaneous Maximum limit is 1100 umhos/cm and reported value was 1207 umhos/cm.	Violation	3	Report

Report displays most recent five years of violations. Refer to the Interactive Violation Report for more data.

Total Violations: 160 **Priority Violations: 0**

*Click the "(+/-) Violation Description" link to expand and contract the violation description.
*As of 5/20/2010, the Water Board's Enforcement Policy requires that all violations be classified as 1, 2 or 3, with class 1 being the highest. Prior to this, violations were simply classified as Yes or No. If a 123 classification has been assigned to a violation that occurred before this date, that classification data will be displayed instead of the Yes/No data.

Violation Types

CAT1 = Category 1 Pollutant (Effluent Violation for Group 1 Pollutant)

Deficient Reporting = Deficient Reporting

OEV = Other Effluent Violation

DMON = Deficient Monitoring

Order Conditions = Order Conditions

Enforcement Actions

Enf Id	Enf Type	Enf Order No.	Effective Date	Status
386614	Oral Communication		07/26/2012	Historical
317357	Notice of Violation		01/05/2006	Historical
251474	Notice of Violation		12/31/2003	Historical
242457	Notice of Violation		06/18/2003	Historical
242108	Notice of Violation		04/30/2002	Historical

Total Enf Actions: 5

Inspections

Inspection ID	Inspection Type	Lead Inspector	Actual End Date	Planned	Violations	Attachment
24456608	B Type compliance inspection	Omar A Mostafa	05/04/2016	N	0	N/A
11847258	B Type compliance inspection	Hossein Aghazeynali	10/24/2012	N	0	N/A
330526	A Type compliance inspection	Amy M Simpson	05/08/2003	Y	0	N/A
300962	A Type compliance inspection	Jeff Gymer	02/20/2002	Y	0	N/A
300967	B Type compliance inspection	Jeff Gymer	04/19/2001	Y	0	N/A
300966	B Type compliance inspection	Kit Lai	06/05/1998	Y	0	N/A
300963	B Type compliance inspection	Martin Sandoval	02/06/1997	Y	0	N/A
300964	A Type compliance inspection	Martin Sandoval	02/01/1995	Y	0	N/A
300965	A Type compliance inspection	Martin Sandoval	03/15/1994	Y	0	N/A

Total Inspections: 9 **Last Inspection: 05/04/2016**

The current report was generated with data as of: 06/03/2016

Appendix B
Well No. 7 Status Summary Letter, February 10, 2016

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February 10, 2016

Betsy Lichti, PE and Sudarshan Poudyal, PE
State Water Resources Control Board
Division of Drinking Water (DDW)
265 W. Bullard Avenue, Suite 101
Fresno, CA 93704

**Subject: Riverdale Public Utility District Arsenic Compliance Project #P84C-1010028-002C &1010028-002C
Well No. 7 Status Summary**

Betsy and Sudarshan:

The purpose of this letter is to summarize action items resulting from our meeting at the DDW Fresno District Office on September 10, 2015 regarding the subject project relating to the implementation of a replacement well for the Riverdale Public Utility District (District). Documentation of action items are as follows:

- 9-10-2015: Meeting was held at the DDW Fresno office to discuss the results of the Riverdale test well at the Well No. 5 site. The test well results showed water quality similar to Well No. 6, including arsenic levels below the MCL. It was decided that the District should pursue a new well project (Well No. 7) to replace Well No. 5, contingent upon approval from the Project Manager at the State Water Board Division of Financial Assistance and RWQCB, in order to avoid constructing a water treatment plant for Well No. 5.

- 10-1-2015: Meeting was held with the RWQCB to discuss elevated EC levels at the District's wastewater treatment plant as a result of source water EC levels from Well No. 6 and the proposed Well No. 7. The RWQCB concurred with the decision to proceed with Well No. 7.

- 10-21-2015: Marques Pitts, Project Manager for the State Water Board Division of Financial Assistance approved (via email) the change order to have Zim Industries construct a new well (Well No. 7) under the test well contract and confirmed with the Environmental Review Unit the need to complete an Addendum to the Mitigated Negative Declaration.

- 11-3-2015: Change Order was executed with Zim Industries to construct Well No. 7 under their existing test well contract.

- 12-7-2015: DDW Fresno office approved (Via Email) the Specifications and Drawings for the well construction.

Zim Industries is tentatively scheduled to start the well construction in February 2016. Provost & Pritchard is currently working on the Plans & Specifications for the Well No. 7 Equipping phase and will submit to DDW for review when complete.

If you have any questions, please do not hesitate to contact me at (559) 449-2700 or kmortensen@ppeng.com.

Sincerely,

Keith Mortensen, P.E.

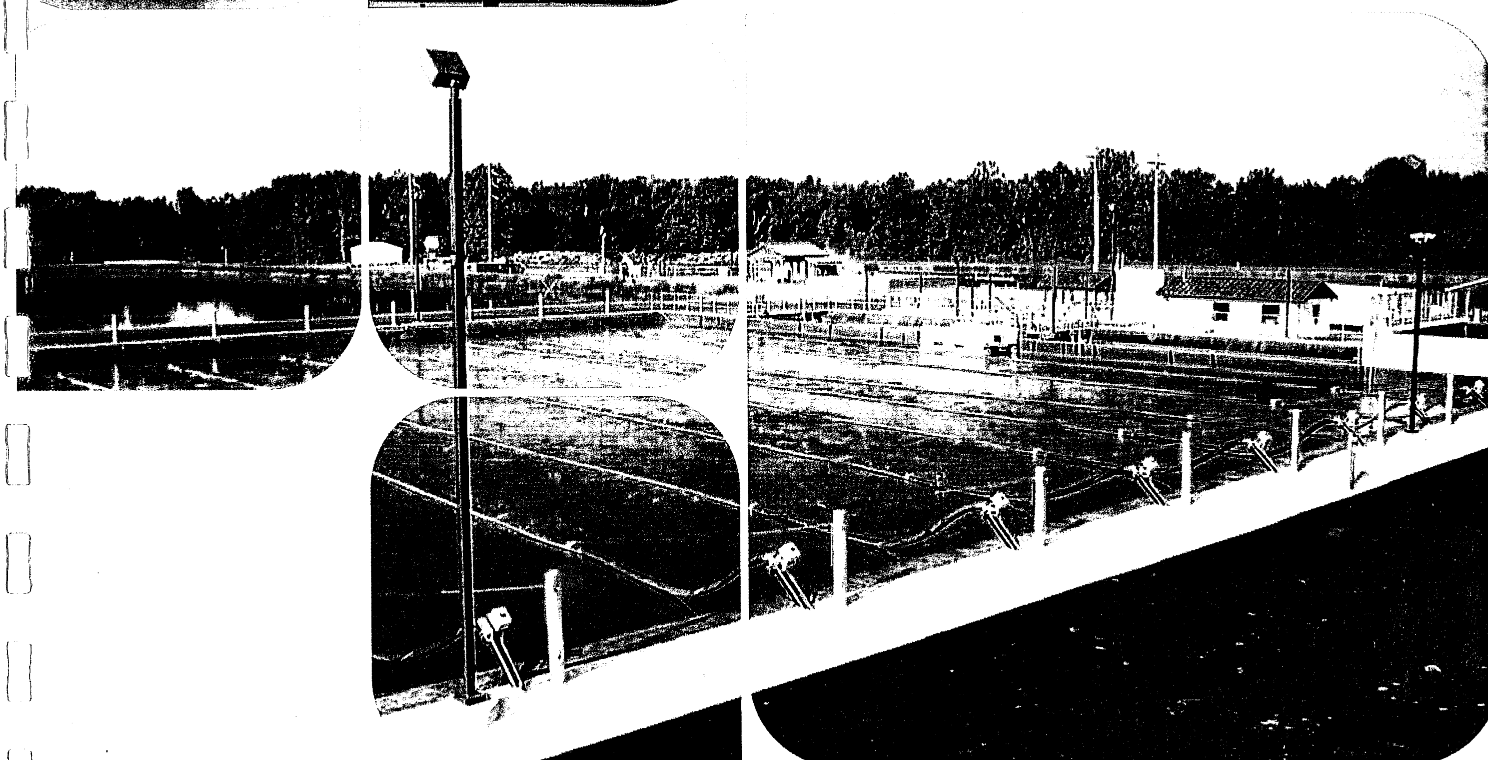
c: Ronald Bass
Riverdale Public Utility District
P.O. Box 248
Riverdale, CA 93656

Marques Pitts
State Water Board – Division of Financial Assistance
Drinking Water State Revolving Fund
1616 Capitol Ave., MS 7418
P.O. Box 997377
Sacramento, CA 95899

Appendix C
Water Balance Calculations

Appendix D
Treatment Process Information

Parkson™
Treating Water Right



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Extended Aeration Treatment System

- Low loaded activated sludge technology
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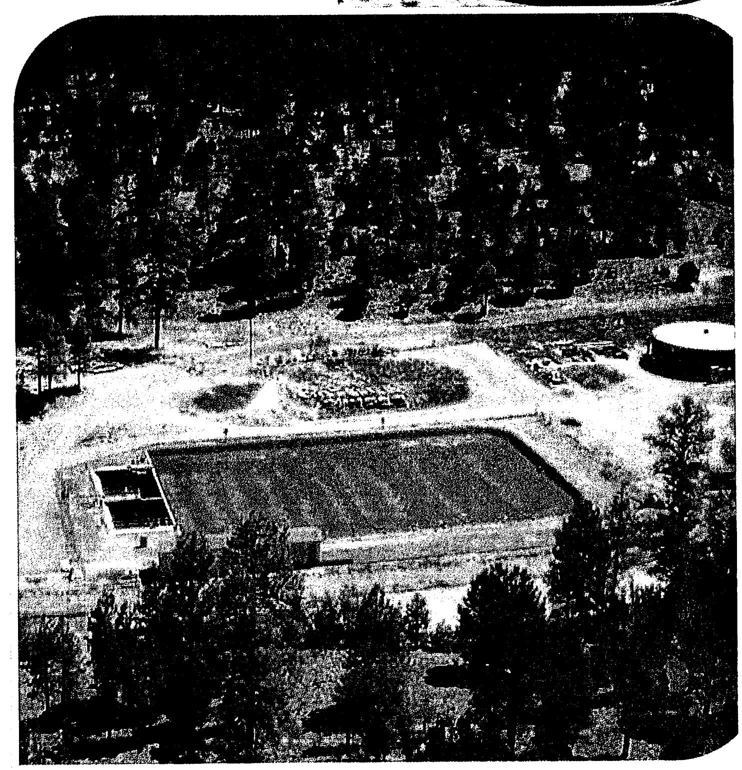
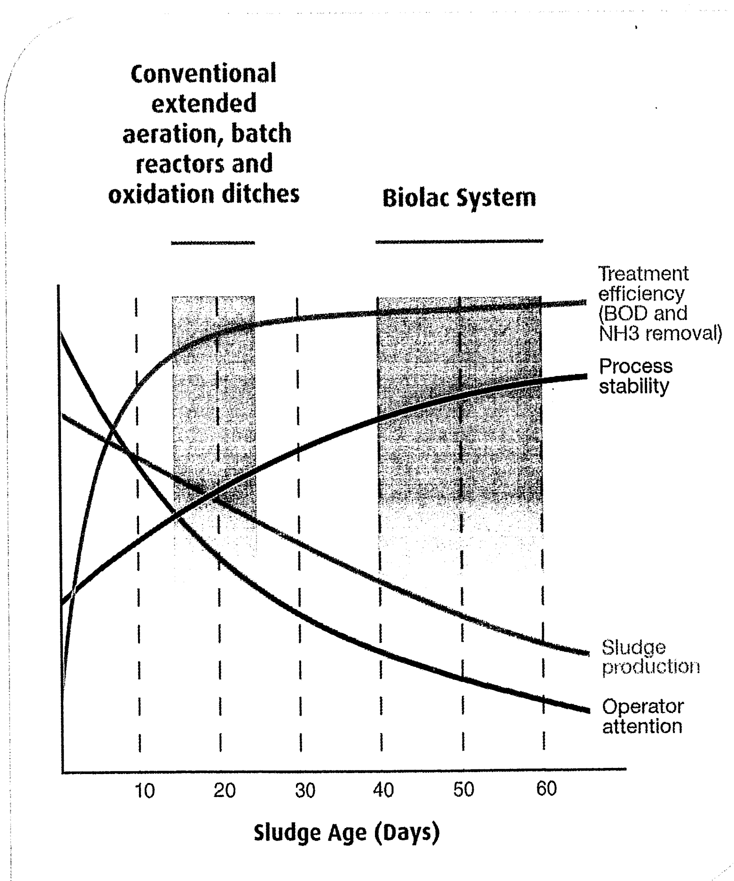
Extended sludge age biological technology



The Biolac® system is an innovative activated sludge process using extended retention of biological solids to create an extremely stable, easily operated system.

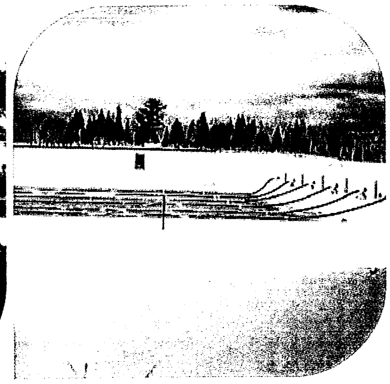
The capabilities of this unique technology far exceed ordinary extended aeration treatment. The Biolac® process maximizes the stability of the operating environment and provides high-efficiency treatment. The design ensures the lowest cost construction and guarantees operational simplicity. Over 700 Biolac® systems are installed throughout North America treating municipal wastewater and many types of industrial wastewater.

The Biolac® system utilizes a longer sludge age than other aerobic systems. Sludge age, also known as SRT (Solids Retention Time) or MCRT (Mean Cell Residence Time), defines the operating characteristics of any aerobic biological treatment system. A longer sludge age dramatically lowers effluent BOD and ammonia levels, especially on colder climates. The Biolac® long sludge age process produces BOD levels of less than 10 mg/L and complete nitrification (less than 1 mg/L ammonia). Minor modifications to the system will extend its capabilities to denitrification and biological phosphorus removal.



While most extended aeration systems reach their maximum mixing capability at sludge ages of approximately 15-25 days, the Biolac® system efficiently and uniformly mixes the aeration volumes associated with a 30-70 day sludge age.

The large quantity of biomass treats widely fluctuating loads with very few operational changes. Extreme sludge stability allows sludge wasting to non-aerated sludge ponds or basins and long storage times.



Aeration components

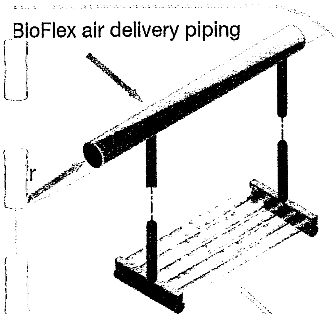
Simple Process Control and Operation

The control and operation of the Biolac® process is similar to that of conventional extended aeration. Parkson provides a very easy to use system to control both the process and aeration. Additional controls required for denitrification, phosphorus removal, dissolved oxygen control and SCADA communications are also easily implemented.

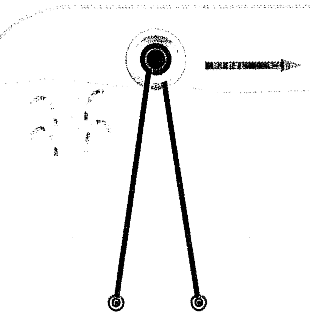
Aeration System Components

The ability to mix large basin volumes using minimal energy is a function of the unique BioFlex moving aeration chains and the attached BioFuser® fine bubble diffuser assemblies. The gentle, controlled, back and forth motion of the chains and diffusers distributes the oxygen transfer and mixing energy evenly throughout the basin area. No additional airflow is required to maintain mixing.

Stationary fine-bubble aeration systems require 8-10 CFM of air per 1000 cu. ft. of aeration basin volume. The Biolac® system maintains the required mixing of the activated sludge and suspension of the solids at only 4 CFM per 1000 cu.ft. of aeration basin volume. Mixing of a Biolac® basin typically requires 35-50 percent of the energy of the design oxygen requirement. Therefore, air delivery to the basin can be reduced during periods of low loading while maintaining effective food to biomass contact and without the risk of solids settling out of the wastewater.



BioFuser fine bubble, air diffuser assembly



Controlled oxygen transfer and mixing movement

System Construction

A major advantage of the Biolac® system is its low installed cost. Most systems require costly in-ground concrete basins for the activated sludge portion of the process. A Biolac® system can be installed in earthen basins, either lined or unlined. The BioFuser® fine bubble diffusers require no mounting to basin floors or associated anchors and leveling. These diffusers are suspended from the BioFlex floating aeration chains; The only concrete structural work required is for the simple internal clarifier(s) and blower/control buildings.

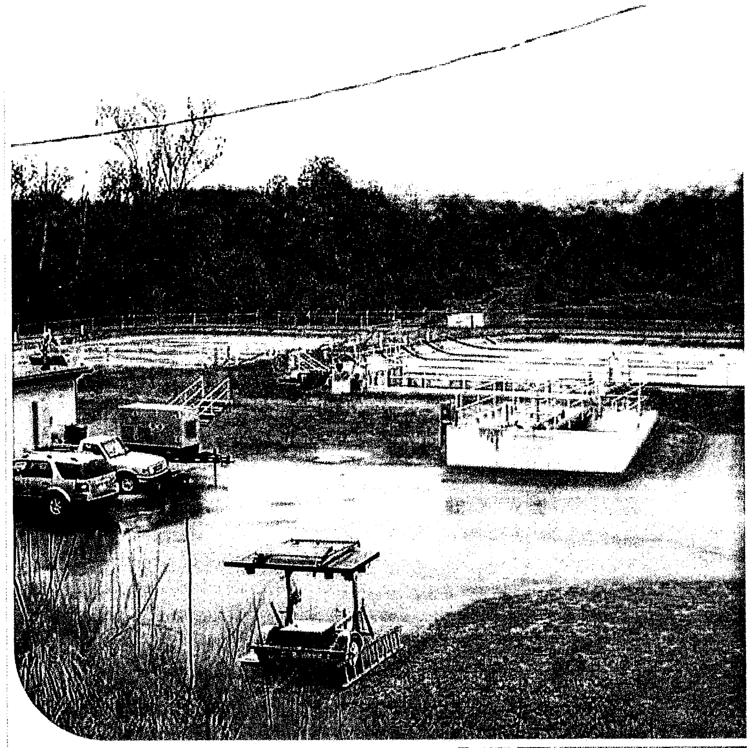
Biological Nutrient Removal

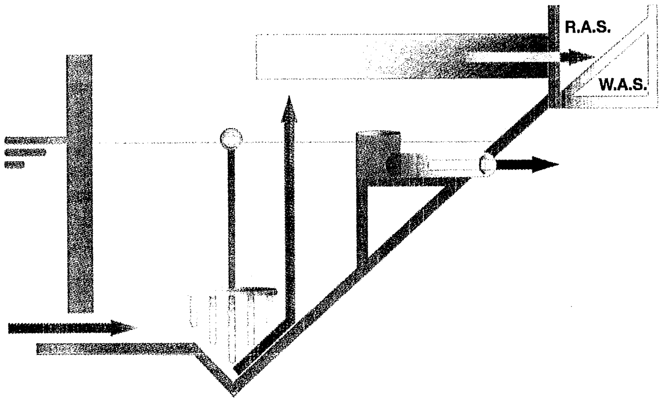
Simple control of the air distribution to the BioFlex® chains creates moving waves of oxic and anoxic zones within the basin. This repeated cycling of environments nitrifies and denitrifies the wastewater without recycled pumping of mixed liquor or additional external basins. This mode of Biolac® operation is known as the Wave Oxidation process. No additional in-basin equipment is required and simple timer-operated actuator valves regulate manipulation of the air distribution.

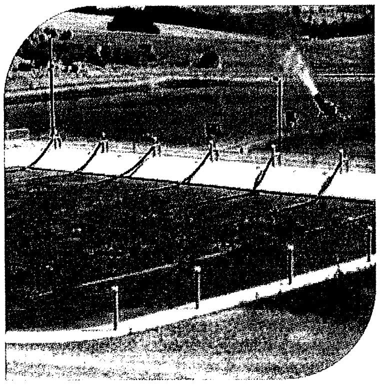
Biological phosphorus removal can also be accomplished by incorporating an anaerobic zone.

Type "R" Clarifier

Land space and hydraulic efficiencies are maximized using the type "R" clarifier. The clarifier design incorporates a common wall between the clarifier and aeration basin. The inlet ports in the bottom of the wall create negligible hydraulic headloss and promote efficient solids removal by filtering the flow through the upper layer of the sludge blanket. The hopper-style bottom simplifies sludge concentration and removal, and minimizes clarifier HRT. The sludge return airlift pump provides important flexibility in RAS flows with no moving parts. All maintenance is performed from the surface without dewatering the clarifier.







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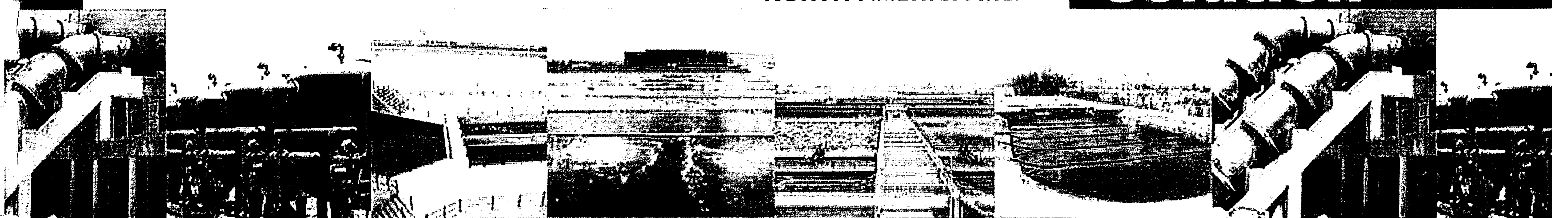
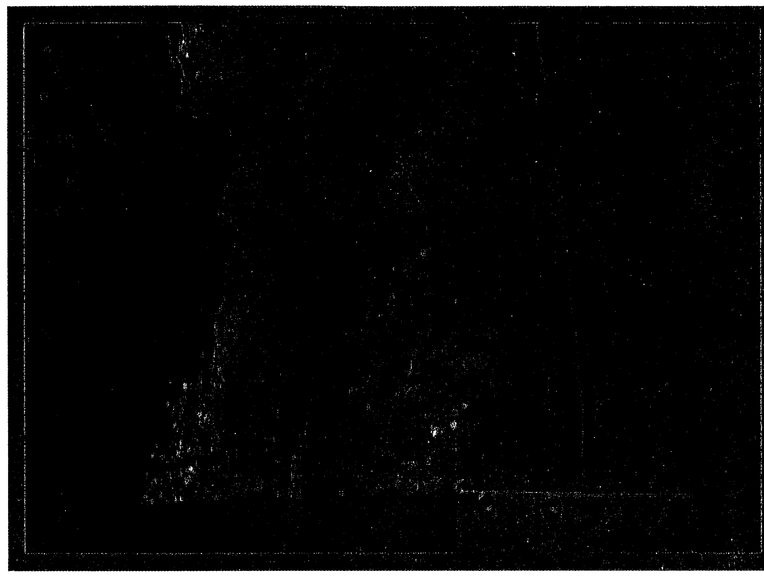
the clear solution



OXIWORKS® HIGH EFFICIENCY FINE BUBBLE DIFFUSER SYSTEM



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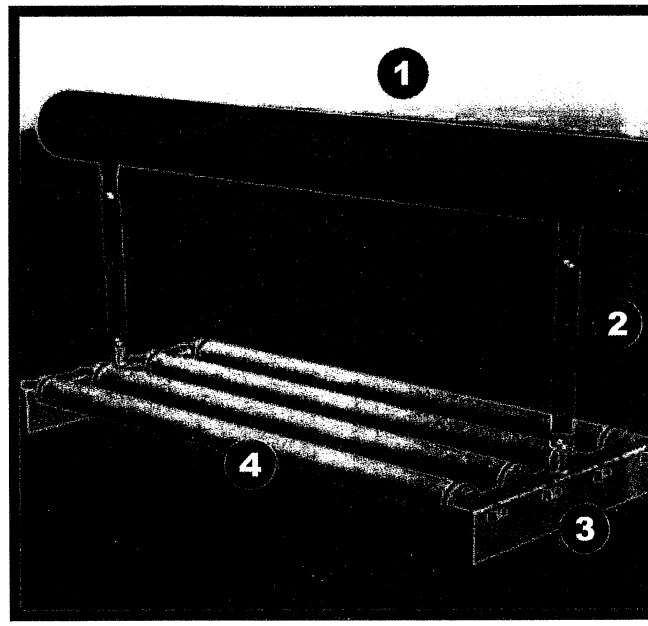
The OXIWORKS® Diffuser

The OXIWORKS® fine bubble diffuser system brings high efficiency aeration to both activated sludge and lagoon treatment systems.



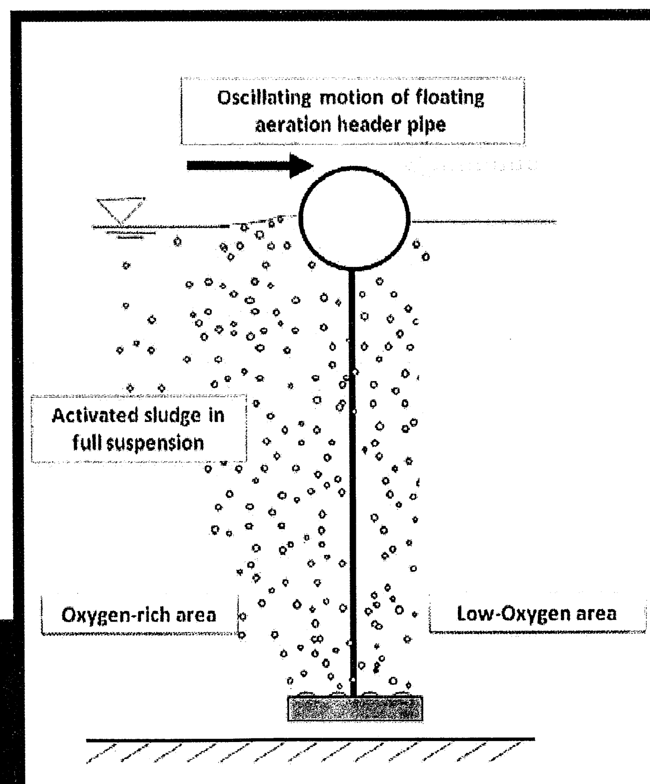
How it works

Air is supplied from the blowers to the air main to each floating lateral header (1). The air then flows through the downcomer hoses (2) into the OXIWORKS® diffuser assembly (3 & 4).

In the diffuser assembly, the air is distributed through high efficiency fine bubble tubular membranes creating 1mm fine bubbles for aeration and mixing.

The system is designed to allow the lateral floating headers with suspended diffusers to move in the basin creating significant advantages:

- The oxygen transfer is maximized because the diffusers are continually move towards areas of low dissolved oxygen. Along with using high efficiency fine bubble diffusers, this increases the overall efficiency of the system.
- Moving diffusers mix a significantly larger area than other systems because the laterals oscillate over a wide area. OXIWORKS® can reduce the required amount of mixing air by 30% compared to fixed aeration systems.



OXIRISE®
Maintenance and Retrieval System

Bioworks has created a simple and effective system to minimize manual labor for system maintenance.

How it Works

The OXIRISE® system allows the operator to raise each diffuser assembly to the surface without ever entering the basin by simply activating a switch at the side of the basin. The benefits of the OXIRISE® system are:

- When work on the diffusers are necessary, they can be retrieved from the surface without reaching under diffuser headers and pulling up of the diffuser assemblies suspended in the basin creating an easier safer way to perform maintenance.
- When the OXIRISE® system is activated periodically, it flexes the fine bubble diffusers, maximizing the life of the membranes.

Applications

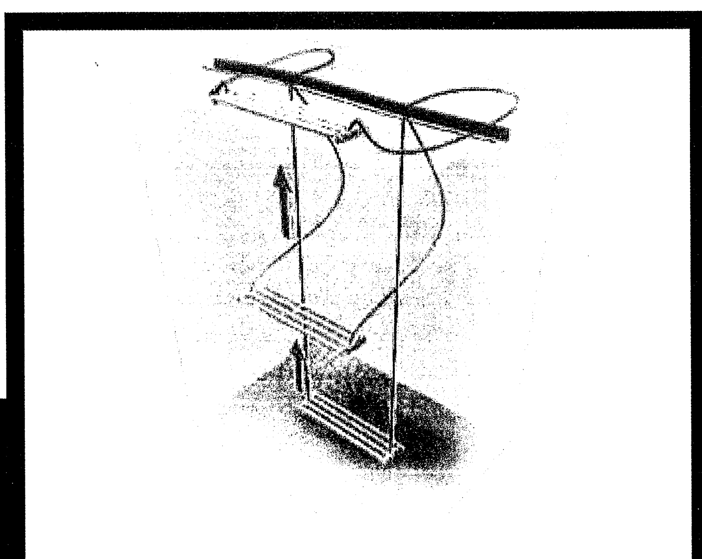
The Bioworks OXIWORKS® moving suspended aeration system utilizing earthen or concrete basins combines a highly stable biological process with minimal investment, operating costs, and maintenance to provide a high quality effluent.

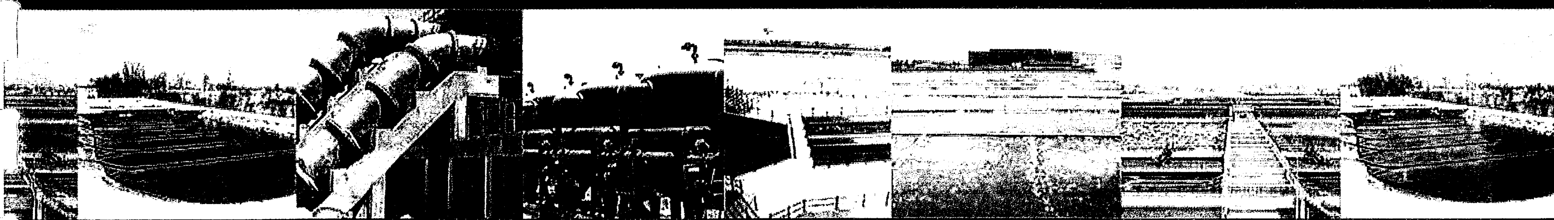
MUNICIPAL APPLICATIONS:

- Aerated Lagoons
- Activated Sludge Systems
- Biological nitrogen and phosphorus removal

INDUSTRIAL APPLICATIONS:

- Food and Meat Processing Plants
- Pulp and Paper Processing Plants
- Industrial Pretreatment Lagoons
- Aerated Lagoons
- Activated Sludge Processes





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Bioworks is a company that helps consulting engineers and customers consistently and professionally meet their objectives: state-of-the-art process design assistance, clean effluent to improve our environment, reduced energy usage, and easy plant operation and maintenance.

Bioworks creates value by designing each project specifically for our customers' needs and our aim is to exceed expectations in terms of design, costs, support, and operation of our systems. We guarantee our process designs and provide industry leading warranties on the equipment we provide.

Bioworks North America Inc. is the US subsidiary of the BIOWORKS group which provides a broad range of wastewater treatment processes worldwide.

Why choose Bioworks?

- We help from the start (the process design) to the finish (installation and then operational support)
- Guaranteed results
- Low energy usage - our diffused aeration system is the finest of its kind
- Simple, low cost operation
- Easy maintenance

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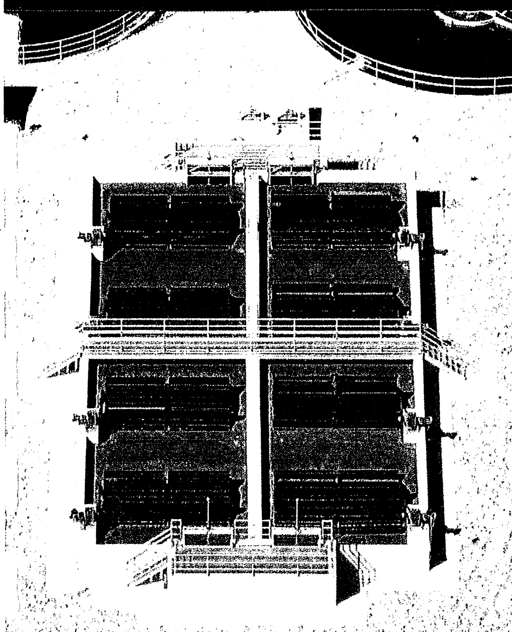
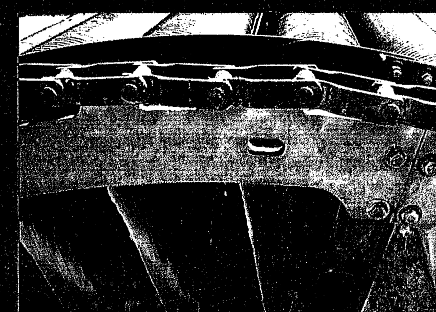
STM-Aerotor™

Advanced Biological Process

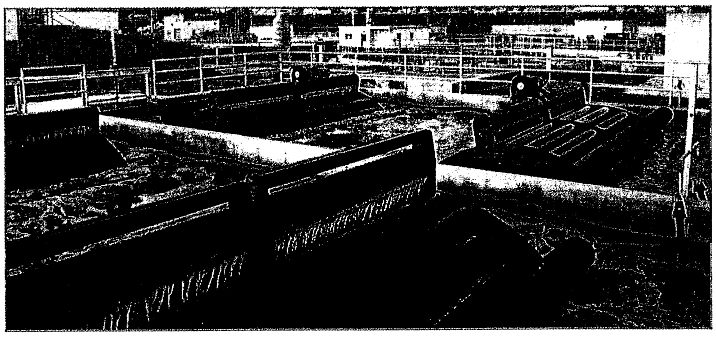




ISO 9001 WesTech CERTIFIED QUALITY SYSTEM

WesTech

STM-Aerotor™



The STM-Aerotor™ biological nutrient removal system uses integrated fixed film and activated sludge technology as part of a process that provides biological nutrient removal for municipal and industrial wastewater treatment.

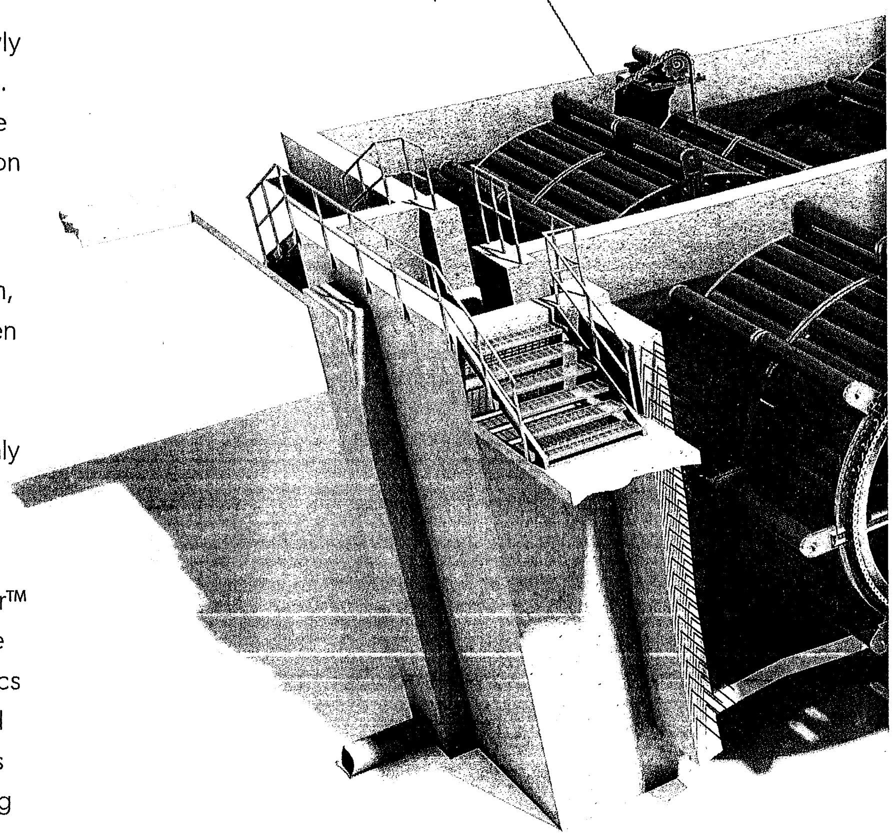
Activated Sludge

The STM-Aerotor™ captures atmospheric air with each rotation, draws it down into the mixed liquor, and slowly releases the compressed air as coarse bubble aeration. The amount of aeration can be controlled using a variable speed drive to speed up or slow down the rotor, based on the actual oxygen demand.

The aeration of the system is achieved without the use of blowers, aeration piping, or diffusers. During the rotation, additional cascade aeration elevates the dissolved oxygen in the upper layer of the basin. The slow rotation of the STM-Aerotor™, intense air release, and the addition of a peripheral mixing paddle combine to ensure a thoroughly mixed system.

Drive Chain

The drive chain is constructed of hardened steel components.

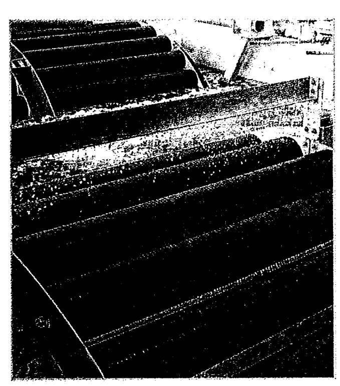


Fixed Film

In addition to ensuring effective aeration, the STM-Aerotor™ also provides a large surface area for fixed film growth. The interior and exterior of the special polypropylene discs provide the perfect environment for a variety of attached growth organisms. The fixed film component increases the "effective" sludge age and improves the sludge settling characteristics. The fixed film will react quickly to an increased food source, or shock load, to eliminate discharge violations during peak or diurnal fluctuations.

Biological Nutrient Removal

The unique STM-Aerotor™ design allows the process tank to develop both anoxic and/or anaerobic zones within the same basin. The process control system can typically achieve total nitrogen below 10mg/L and total phosphorous below 2 mg/L. For enhanced biological nutrient removal, the STM-Aerotor™ system can be combined with additional biological selector zones in separate basins to achieve total nitrogen below 3 mg/L and total phosphorus below 1 mg/L.



Mixing Paddle

To allow the system to be deep enough to develop zones that are oxygen limited, a mixing paddle is added to ensure mixing.

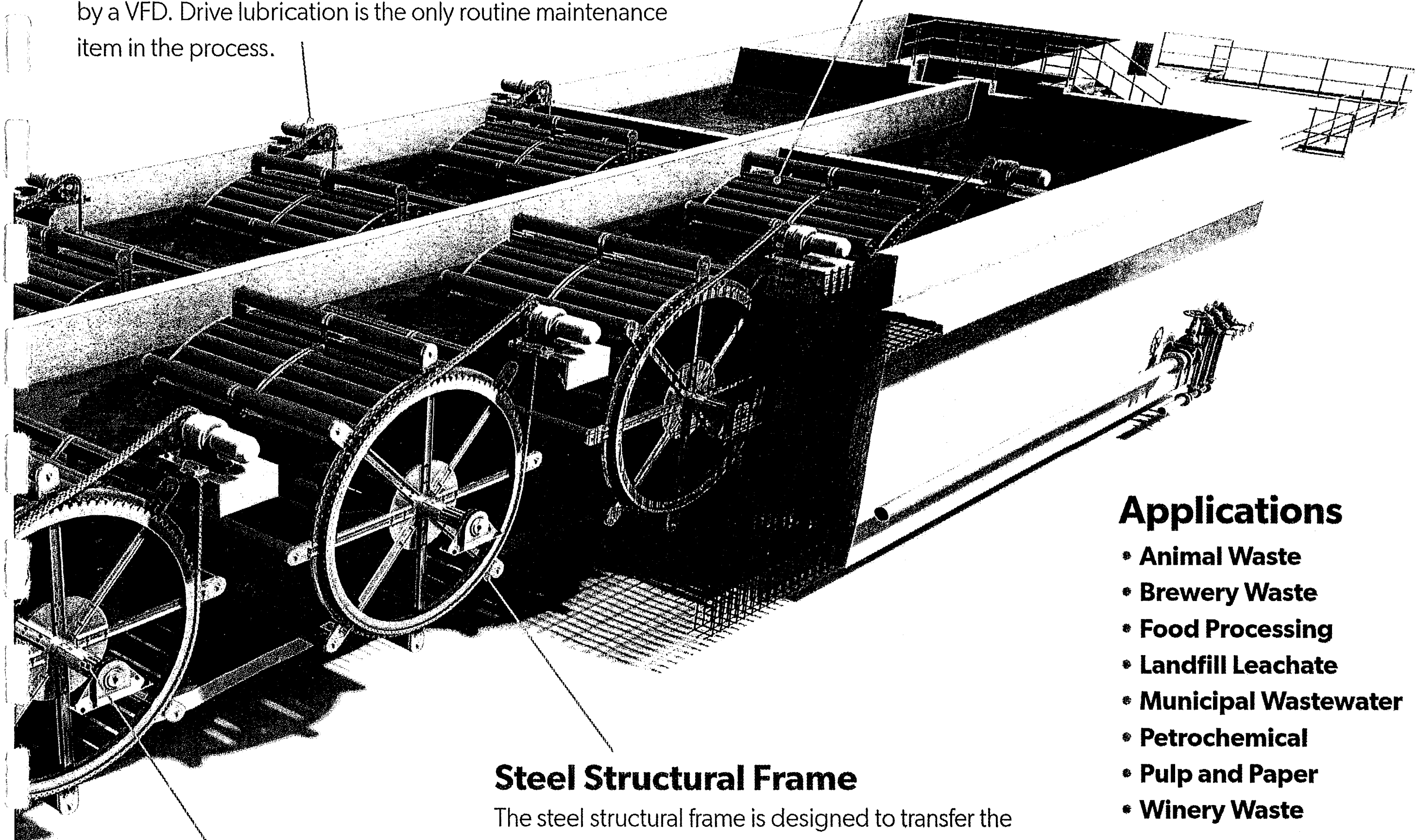
Integrated Fixed Film and Activated Sludge

Media

The special STM-Aerotor™ media discs maximize the surface area for the fixed film, while optimizing the captured air volume and release depth for efficient aeration.

Low-HP Drive Unit

The simple drive unit is easily accessible and is controlled by a VFD. Drive lubrication is the only routine maintenance item in the process.



Steel Structural Frame

The steel structural frame is designed to transfer the loads from the media to the supports at the wall.

Steel Center Shaft

The center shaft provides connecting points for the STM-Aerotor™ frame assembly.

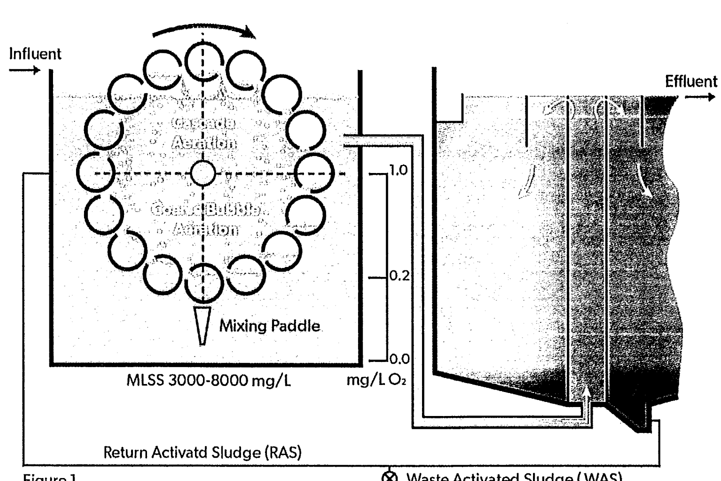


Figure 1

Applications

- Animal Waste
- Brewery Waste
- Food Processing
- Landfill Leachate
- Municipal Wastewater
- Petrochemical
- Pulp and Paper
- Winery Waste

Features

- Compact Design
- Low-HP Drives
- No Blowers, Aeration Piping, or Diffusers
- Peripheral Mixing Paddle
- Rigid Structural Frame
- UV-protected Media

Benefits

- Advanced Biological Nutrient Removal
- Improved Sludge Quality
- Low Capital Cost
- Low Energy Usage
- Minimized Operator Attention
- No Noise or Odor Problems
- Simple Construction
- Stable Process Performance

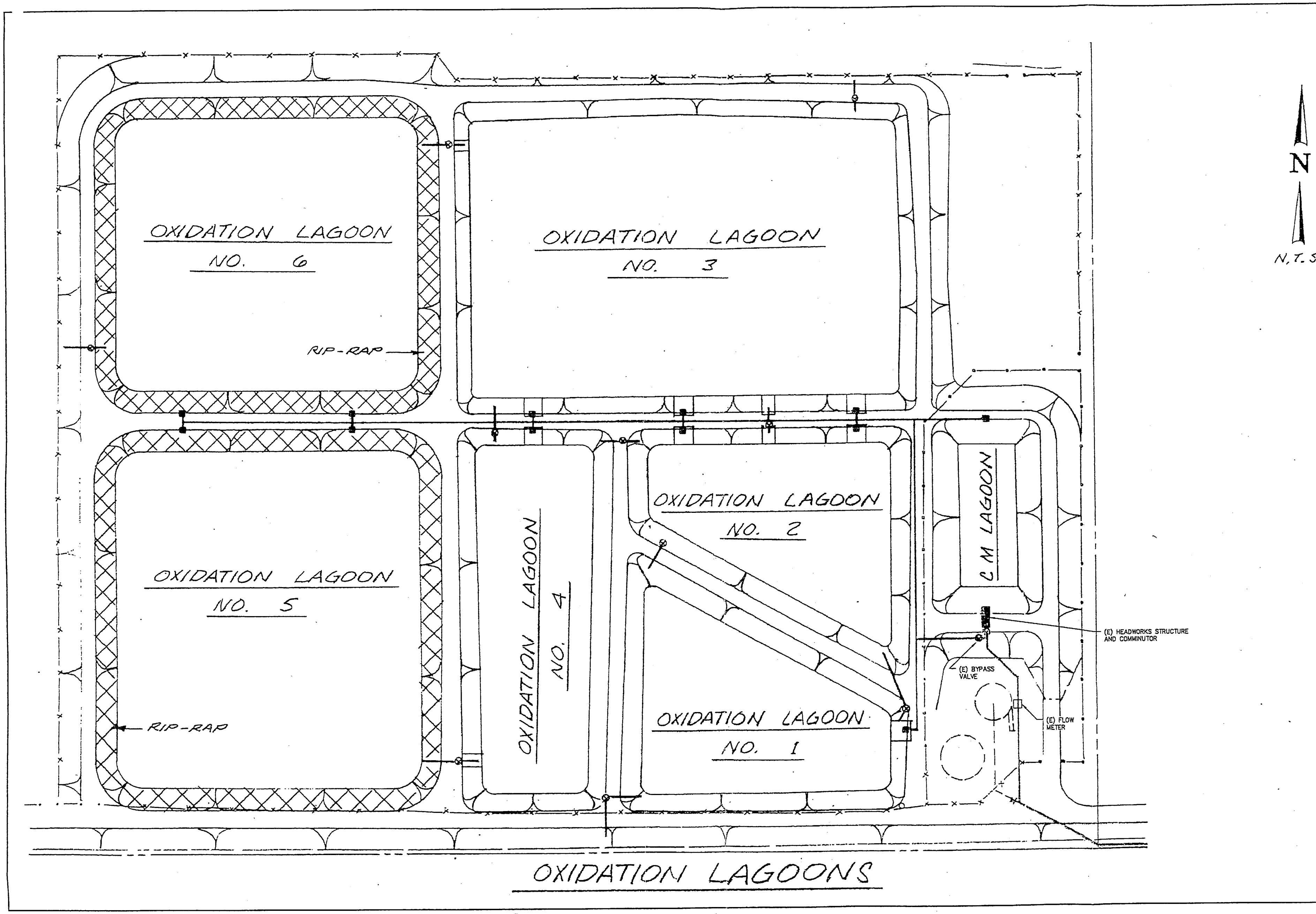


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FIGURE 3-1

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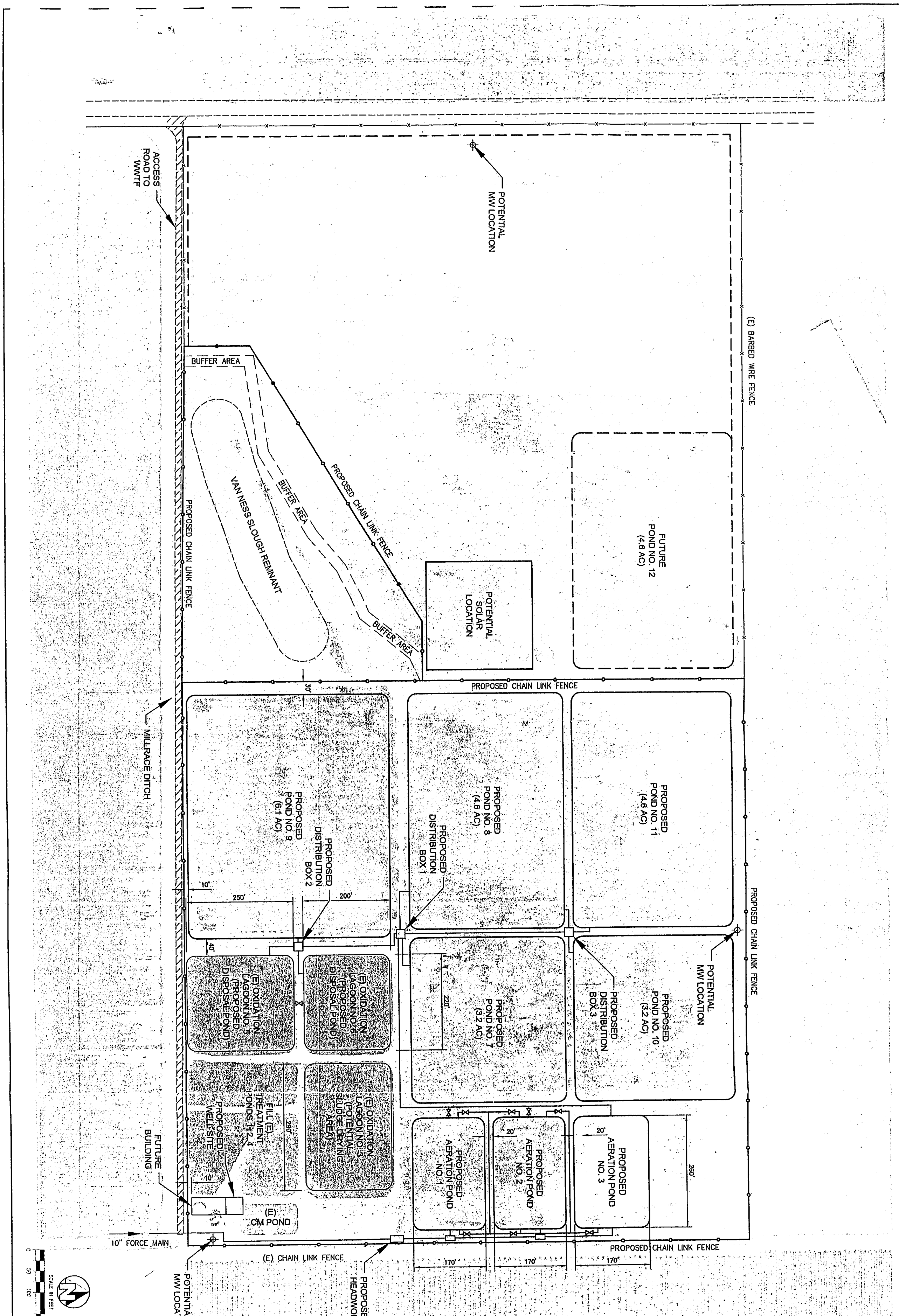

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FIGURE 5-1
PROPOSED STORAGE AND DISPOSAL PONDS

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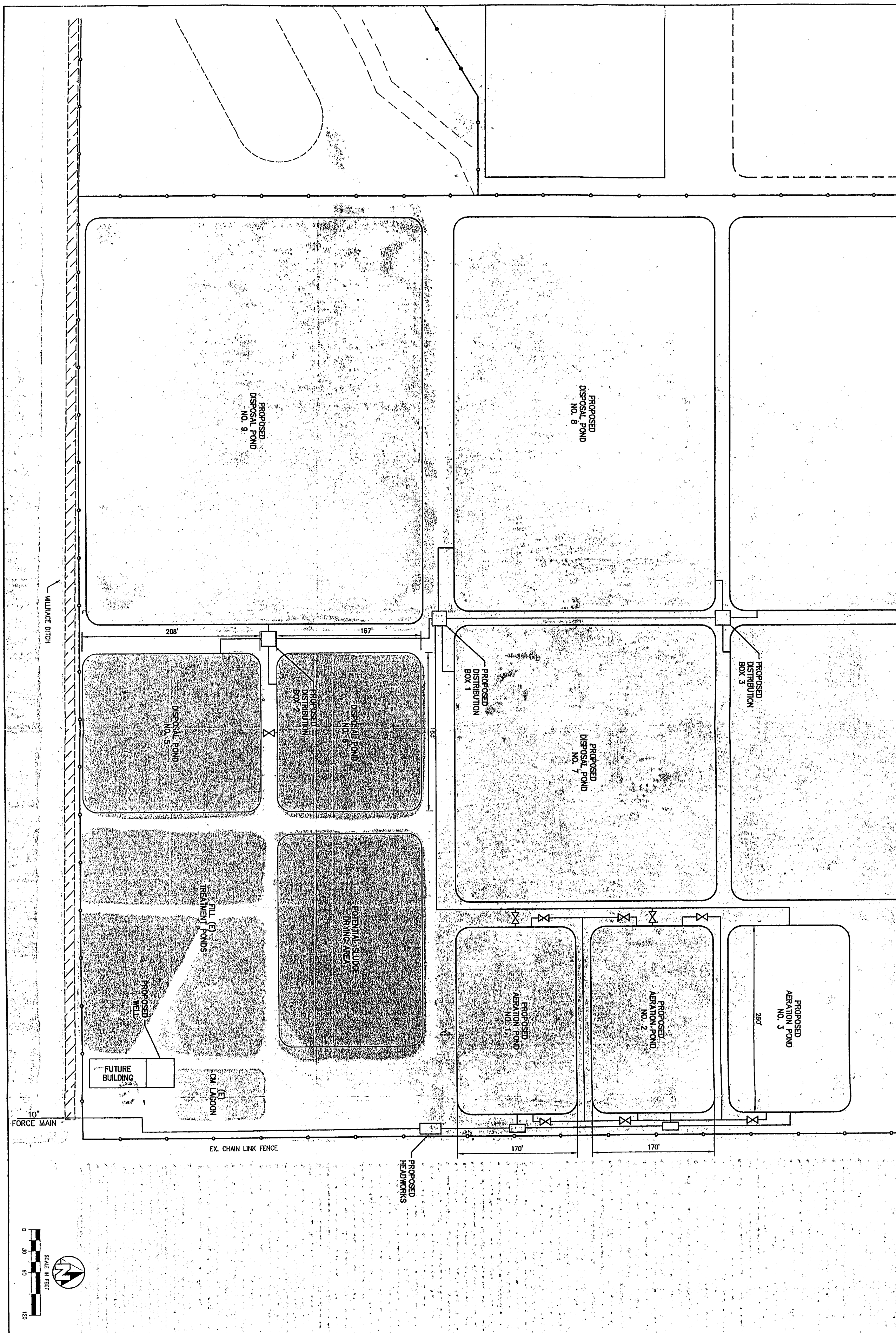

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FIGURE 6-1

AERATION POND SYSTEM

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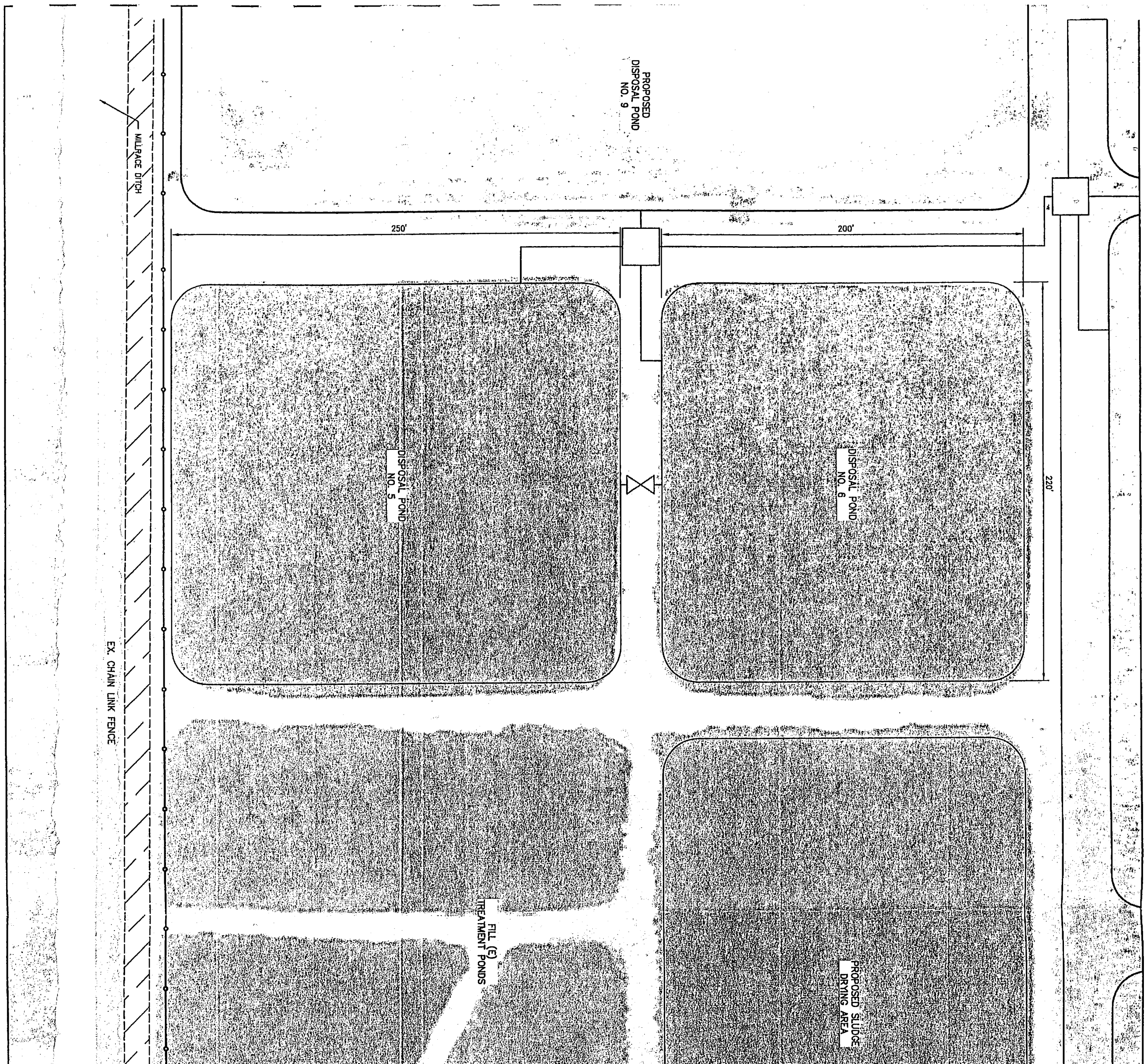

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FIGURE 6-2

BIOLAC PROCESS

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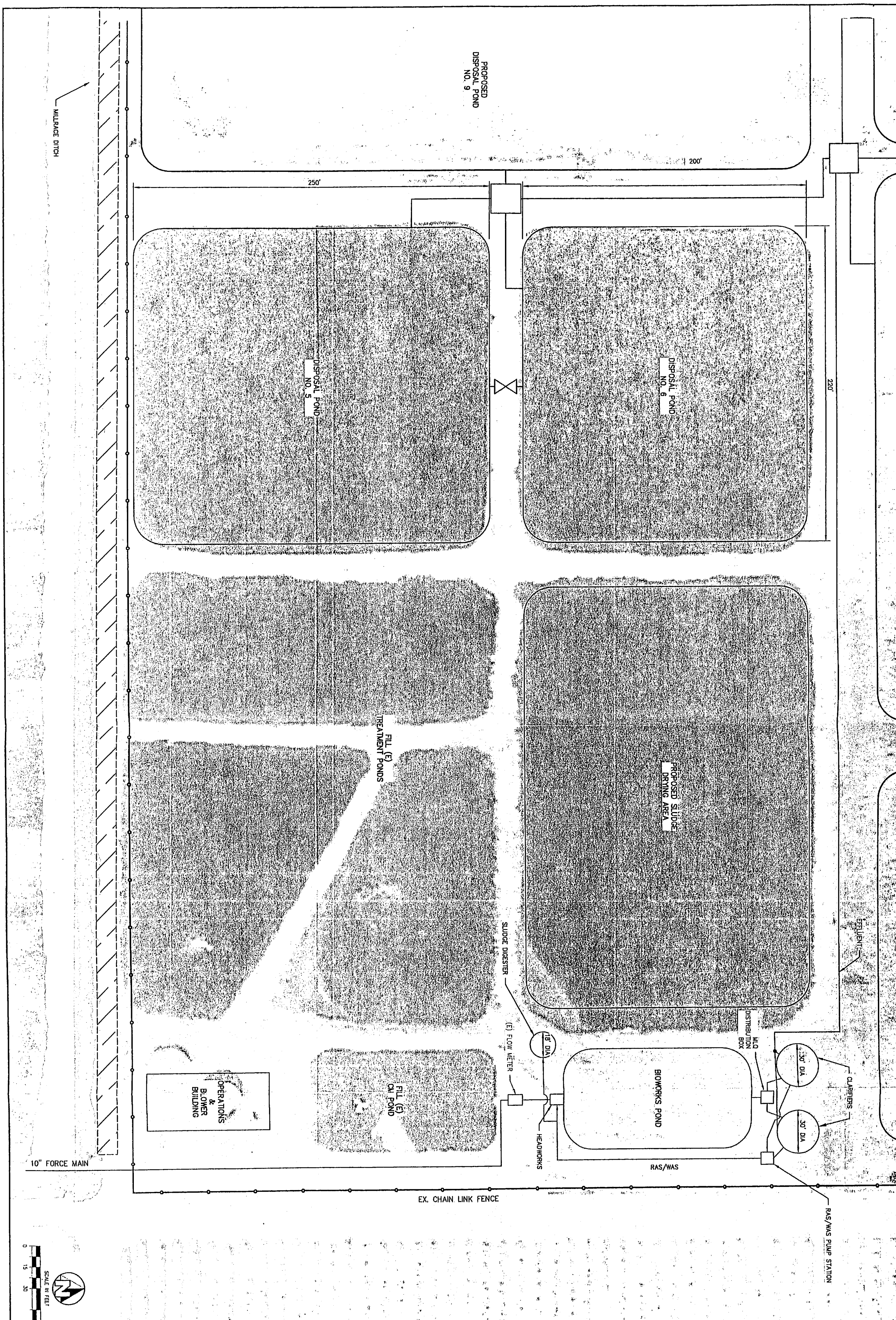

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FIGURE 6-3

BIOWORKS PROCESS

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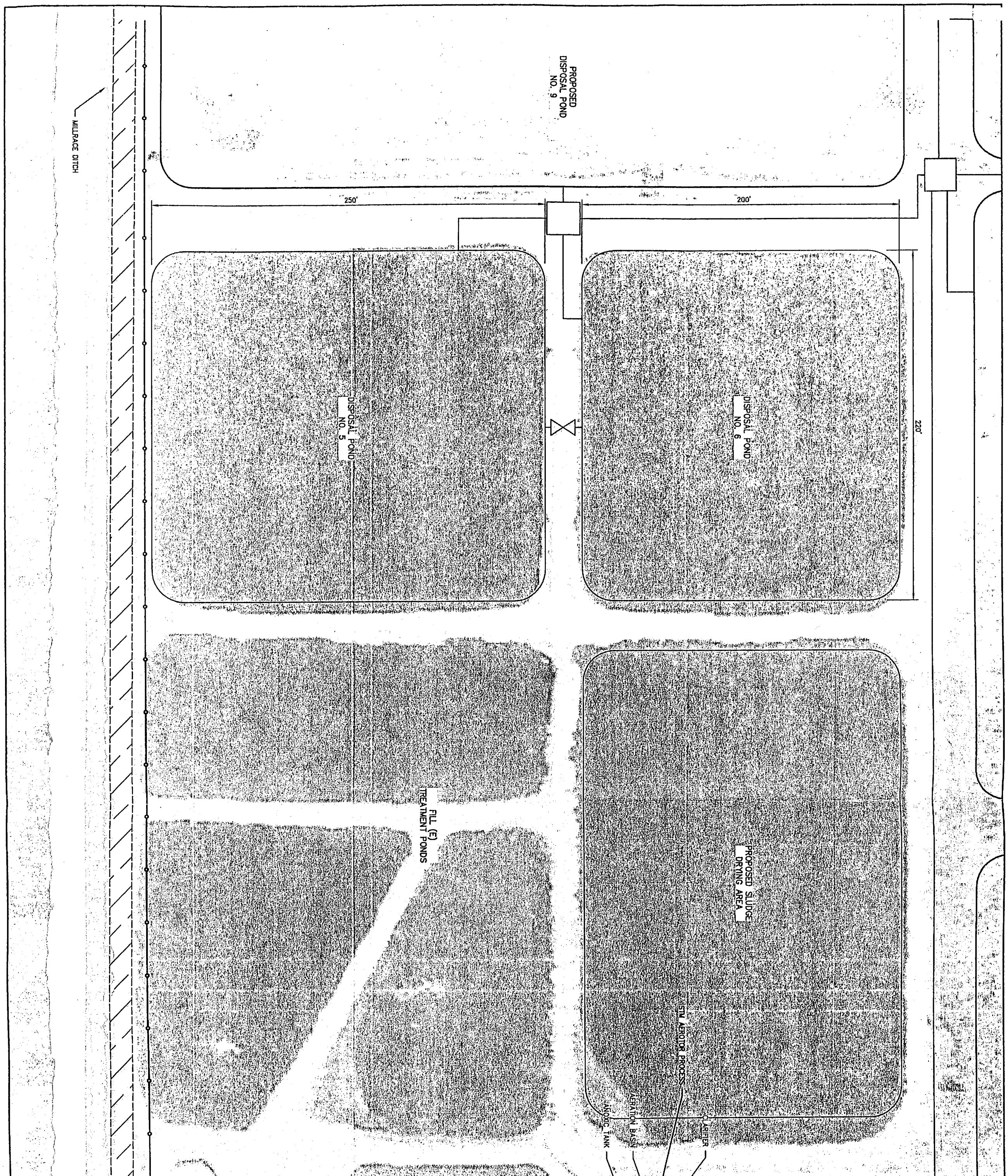

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FIGURE 6-4

STM AERORATERS PROCESS

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FIGURE 6–5

MBR PROCESS

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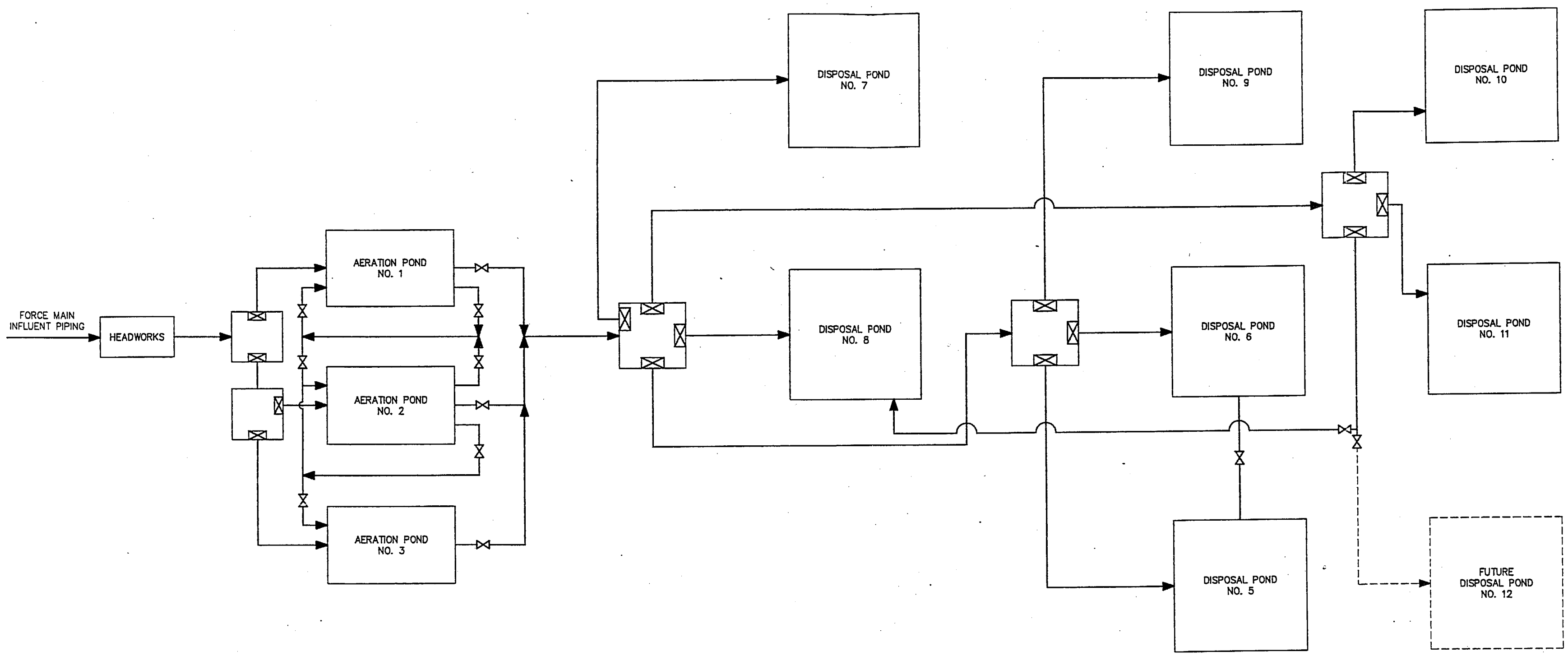

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FIGURE 7-1

PRELIMINARY PROCESS FLOW DIAGRAM

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SHEET 1 OF —

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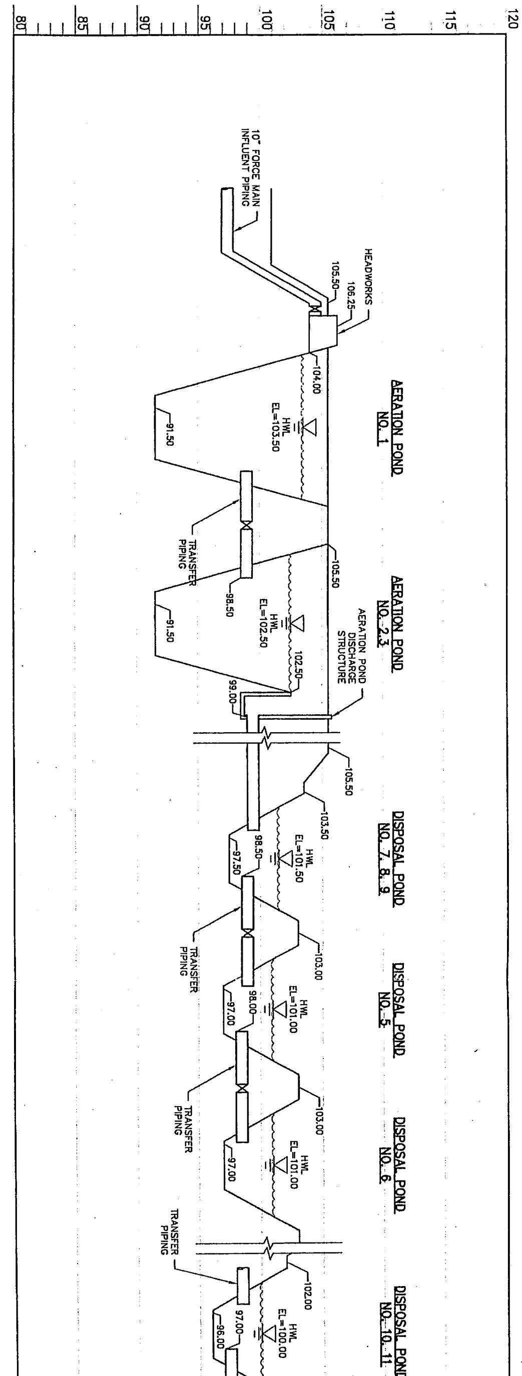

WARNING POWER LINES OVERHEAD

811. Know what's below. Call before you dig.

DESIGN ENGINEER:	
LICENSE NO:	
DRAFTED BY:	CHECKED BY:
DATE: 1-24-17	
JOB NO: 22111600	
PROJECT NO:	
PHASE:	

ORIGINAL SCALE SHOWN IS ONE INCH. ADJUST SCALE FOR REDUCED OR ENLARGED PLANS.

SHEET 1 OF 1

EST. 1968 PROVOST & PRITCHARD CONSULTING GROUP
An Employee Owned Company
286 WEST CROMWELL AVENUE
FRESNO, CALIFORNIA 93711-6162
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WWTP IMPROVEMENTS
PRELIMINARY ENGINEERING REPORT
RIVERDALE PUBLIC UTILITY DISTRICT
RIVERDALE, CA

FIGURE 7-2

PRELIMINARY HYDRAULIC PROFILE

FOR REVIEW ONLY

PRELIMINARY
NOT FOR CONSTRUCTION
1-24-17

No.	REVISION	BY	DATE

1/24/2017 12:04 PM G:\Riverdale PUD - 2211\221116001-Wastewater Treatment Plant\DAC\EXHIBIT\I PRELIMINARY HYDRAULIC PROFILE.dwg -Stephen Grisez

Riverdale Public Utility District

Wastewater Treatment Plant Disposal

Normal Year Rainfall - 0.275 MGD

DATA:

Month	Number of Days per Month	Working Days per Month	Normal Year Rainfall (in/month)	Normal Year Evaporation (in/month)	ET_c Alfalfa (in/month)
January	31	31	2.02	0.91	0.79
February	28	28	1.83	2.65	2.07
March	31	31	1.84	5.01	3.89
April	30	30	1.03	6.95	4.74
May	31	31	0.36	9.15	6.15
June	30	30	0.15	9.48	6.46
July	31	31	0.01	9.98	6.61
August	31	31	0.01	8.45	5.49
September	30	30	0.15	6.74	4.52
October	31	31	0.51	4.34	1.98
November	30	30	1.12	1.31	0.80
December	31	31	1.65	1.24	1.07
Total	**365**	**365**	**10.68**	**66.20**	**44.57**

Effluent Production =	275,000	gpd
Treatment Pond (Lined)	1.80	acres
Approx. Storage Volume =	[illegible]	ac-ft
Approx. Storage Volume =	[illegible]	gal
Pond Percolation Rate =	0.00	in/day
Existing Ponds 5 and 6	2.41	acres
Approx. Storage Volume =	[illegible]	ac-ft
Approx. Storage Volume =	[illegible]	gal
Pond Percolation Rate =	0.33	in/day
Additional Ponds	21.7	acres
Approx. Storage Volume =	[illegible]	ac-ft
Approx. Storage Volume =	[illegible]	gal
Pond Percolation Rate =	0.33	in/day

Alfalfa Area (Available) =	0	acres
Alfalfa Area (needed) =	**0**	acres
Alfalfa Rootzone AWHC =	3.00	inch

STORAGE POND CALCULATIONS:

Month	Effluent Produced (gal/month)	Effluent to Alfalfa (gal/month)	Effluent to Ponds (gal/month)	Surface Rainfall (gal/month)	Surface Evaporation (gal/month)	Pond Percolation (gal/month)	Monthly Available (gal/month)	Cumulative Available (gal/month)
January	8,525,000	0	8,525,000	1,421,207	642,005	6,697,477	2,606,725	8,048,085
February	7,700,000	0	7,700,000	1,287,529	1,864,454	6,049,334	1,073,741	9,121,826
March	8,525,000	0	8,525,000	1,294,564	3,526,633	6,697,477	-404,546	8,717,280
April	8,250,000	0	8,250,000	724,675	4,889,795	647,874	3,437,006	12,154,286
May	8,525,000	0	8,525,000	253,284	6,437,643	6,697,477	-4,356,836	7,797,450
June	8,250,000	0	8,250,000	105,535	6,666,303	6,481,430	-4,792,198	3,005,252
July	8,525,000	0	8,525,000	7,036	7,018,087	4,519,201	-3,005,252	0
August	8,525,000	0	8,525,000	7,036	5,945,146	2,586,890	0	0
September	8,250,000	0	8,250,000	105,535	4,740,287	3,615,248	0	0
October	8,525,000	0	8,525,000	358,819	495,861	6,697,477	1,690,481	1,690,481
November	8,250,000	0	8,250,000	787,996	923,432	6,481,430	1,633,134	3,323,615
December	8,525,000	0	8,525,000	1,160,887	870,665	6,697,477	2,117,745	5,441,360
Total	**100,375,000**	**0**	**100,375,000**	**7,514,103**	**44,020,311**	**63,868,792**	**0**	*Start at 0 Stored October 1st*
ac-ft	**308**	**0**	**308**	**23**	**135**	**196**		

RECLAMATION AREA:

Month	Alfalfa Effluent Applied (gal/month)	Effluent Applied (in)	Effective Rainfall (in)	Fresh Irrigation (in)	Gross Crop Need (in)	Soil Moisture Start (in)	Soil Moisture End (in)	Percolation & Leaching > 3.00 in
		0.0 acres			Irrigation Application Efficiency =			70%
January	0	#DIV/0!	1.36	0.00	1.13	#DIV/0!	#DIV/0!	#DIV/0!
February	0	#DIV/0!	1.18	0.00	2.96	#DIV/0!	#DIV/0!	#DIV/0!
March	0	#DIV/0!	0.47	0.00	5.56	#DIV/0!	#DIV/0!	#DIV/0!
April	0	#DIV/0!	0.00	0.00	6.77	#DIV/0!	#DIV/0!	#DIV/0!
May	0	#DIV/0!	0.00	0.00	8.79	#DIV/0!	#DIV/0!	#DIV/0!
June	0	#DIV/0!	0.00	0.00	9.23	#DIV/0!	#DIV/0!	#DIV/0!
July	0	#DIV/0!	0.00	0.00	9.44	#DIV/0!	#DIV/0!	#DIV/0!
August	0	#DIV/0!	0.00	0.00	7.84	#DIV/0!	#DIV/0!	#DIV/0!
September	0	#DIV/0!	0.00	0.00	6.46	#DIV/0!	#DIV/0!	#DIV/0!
October	0	#DIV/0!	0.26	0.00	2.83	3.00	#DIV/0!	#DIV/0!
November	0	#DIV/0!	0.51	0.00	1.14	#DIV/0!	#DIV/0!	#DIV/0!
December	0	#DIV/0!	1.01	0.00	1.53	#DIV/0!	#DIV/0!	#DIV/0!
	0	**#DIV/0!**	**4.79**	**0.00**	**63.68**			**#DIV/0!**
	0.0	ac-ft	**0.0**	**0.0**	**0.0**	ac-ft		**#DIV/0!**
Percent of Total =	**#DIV/0!**		**#DIV/0!**	**#DIV/0!**				

Total Pond Storage Available (gal)	*33,800,000*
Cumulative Storage Needed (gal)	*12,154,286*
Check	*OK*

Assumptions

1. Rainfall Data per the Western Regional Climate Center.
2. Evaporation data based on 125% of Evapotranspiration (Eto), based on DWR correspondence.
3. Evapotranspiration per the CalPoly Irrigation Training & Research Center (ITRC).
4. Percolation rates are a conservative estimate based on soil types in the area.

EST. 1968 PROVOST & PRITCHARD ENGINEERING GROUP

Riverdale Public Utility District
Wastewater Treatment Plant Disposal
Wet Year Rainfall - 0.275 MGD

DATA:

Month	Number of Days per Month	Working Days per Month	Wet Year Rainfall (in/month)	Wet Year Evaporation (in/month)	ET_c Alfalfa (in/month)
January	31	31	5.14	0.49	0.41
February	28	28	3.70	1.01	0.88
March	31	31	4.53	3.45	2.97
April	30	30	2.76	5.15	4.49
May	31	31	0.01	5.10	4.47
June	30	30	0.00	7.89	5.89
July	31	31	0.00	9.36	6.19
August	31	31	0.09	8.75	5.74
September	30	30	1.03	5.98	3.90
October	31	31	0.09	4.35	1.90
November	30	30	2.51	1.31	1.06
December	31	31	1.75	1.28	1.10
Total	**365**	**365**	**21.61**	**54.11**	**39.00**

Effluent Production =	275,000	gpd
Treatment Pond (Lined)	**1.80**	acres
Approx. Storage Volume =	[illegible]	ac-ft
Approx. Storage Volume =	[illegible]	gal
Pond Percolation Rate =	**0.00**	in/day
Existing Ponds 5 and 6	**2.41**	acres
Approx. Storage Volume =	[illegible]	ac-ft
Approx. Storage Volume =	[illegible]	gal
Pond Percolation Rate =	**0.33**	in/day
Additional Ponds	**21.7**	acres
Approx. Storage Volume =	[illegible]	ac-ft
Approx. Storage Volume =	[illegible]	gal
Pond Percolation Rate =	**0.33**	in/day

Alfalfa Area (Available) =	0	acres
Alfalfa Area =	**0**	acres
Alfalfa Rootzone AWHC =	**3.00**	inch

STORAGE POND CALCULATIONS:

Month	Effluent Produced (gal/month)	Effluent to Alfalfa (gal/month)	Effluent to Ponds (gal/month)	Surface Rainfall (gal/month)	Surface Evaporation (gal/month)	Pond Percolation (gal/month)	Monthly Available (gal/month)	Cumulative Available (gal/month)
January	8,525,000	0	8,525,000	3,616,337	342,989	6,697,477	5,100,871	9,873,681
February	7,700,000	0	7,700,000	2,603,200	712,362	6,049,334	3,541,504	13,415,185
March	8,525,000	0	8,525,000	3,187,161	2,427,308	6,697,477	2,587,376	16,002,561
April	8,250,000	0	8,250,000	1,941,847	3,623,373	6,481,430	87,044	16,089,605
May	8,525,000	0	8,525,000	7,036	3,588,195	6,697,477	-1,753,636	14,335,969
June	8,250,000	0	8,250,000	0	5,549,389	6,481,430	-3,780,819	10,555,150
July	8,525,000	0	8,525,000	0	6,587,152	6,697,477	-4,759,629	5,795,521
August	8,525,000	0	8,525,000	63,321	6,156,216	6,697,477	-4,265,372	1,530,149
September	8,250,000	0	8,250,000	724,675	4,203,816	6,301,008	-1,530,149	0
October	8,525,000	0	8,525,000	63,321	3,060,519	5,527,802	0	0
November	8,250,000	0	8,250,000	1,765,955	923,432	6,481,430	2,611,093	2,611,093
December	8,525,000	0	8,525,000	1,231,243	897,049	6,697,477	2,161,717	4,772,810
Total	**100,375,000**	**0**	**100,375,000**	**15,204,096**	**38,071,800**	**77,507,296**	**0**	*Start at 0 Stored October 1st*
ac-ft	**308**	**0**	**308**	**47**	**117**	**238**		

Assumptions
1. Rainfall Data per the Western Regional Climate Center.
2. Evaporation data based on 125% of Evapotranspiration (Eto), based on DWR correspondence.
3. Evapotranspiration per the CalPoly Irrigation Training & Research Center (ITRC).
4. Percolation rates are a conservative estimate based on soil types in the area.

Total Pond Storage Available (gal)	*33,800,000*
Cumulative Storage Needed (gal)	*16,089,605*
Check	*OK*

Note: Sufficient storage is available; cumulative volume after wet year could be stored over and evaporated and percolated during a normal year.

RECLAMATION AREA:

Alfalfa 0.0 acres — **Irrigation Application Efficiency =** 70%

Month	Effluent Applied (gal/month)	Effluent Applied (in)	Effective Rainfall (in)	Fresh Irrigation (in)	Gross Crop Need (in)	Soil Moisture Start (in)	Soil Moisture End (in)	Percolation & Leaching >3.00 in
January	0	#DIV/0!	4.29	0.00	0.59	#DIV/0!	#DIV/0!	#DIV/0!
February	0	#DIV/0!	2.94	0.00	1.26	#DIV/0!	#DIV/0!	#DIV/0!
March	0	#DIV/0!	2.72	0.00	4.24	#DIV/0!	#DIV/0!	#DIV/0!
April	0	#DIV/0!	0.00	0.00	6.41	#DIV/0!	#DIV/0!	#DIV/0!
May	0	#DIV/0!	0.00	0.00	6.39	#DIV/0!	#DIV/0!	#DIV/0!
June	0	#DIV/0!	0.00	0.00	8.41	#DIV/0!	#DIV/0!	#DIV/0!
July	0	#DIV/0!	0.00	0.00	8.84	#DIV/0!	#DIV/0!	#DIV/0!
August	0	#DIV/0!	0.00	0.00	8.20	#DIV/0!	#DIV/0!	#DIV/0!
September	0	#DIV/0!	0.00	0.00	5.57	#DIV/0!	#DIV/0!	#DIV/0!
October	0	#DIV/0!	0.00	0.00	2.71	3.00	#DIV/0!	#DIV/0!
November	0	#DIV/0!	1.82	0.00	1.51	#DIV/0!	#DIV/0!	#DIV/0!
December	0	#DIV/0!	1.11	0.00	1.57	#DIV/0!	#DIV/0!	#DIV/0!
	0	**#DIV/0!**	**12.88**	**0.00**	**55.70**			**#DIV/0!**
	0.0 ac-ft		**0.0**	**0.0**	**0.0** ac-ft			**#DIV/0!**
Percent of Total =	**#DIV/0!**		**#DIV/0!**	**#DIV/0!**				

EST. 1968 PROVOST & PRITCHARD ENGINEERING GROUP

Riverdale Public Utility District
Wastewater Treatment Plant Disposal
Wet Year Rainfall - 0.325 MGD

DATA:

Month	Number of Days per Month	Working Days per Month	Wet Year Rainfall (in/month)	Wet Year Evaporation (in/month)	ET_c Alfalfa (in/month)
January	31	31	5.14	0.49	0.41
February	28	28	3.70	1.01	0.88
March	31	31	4.53	3.45	2.97
April	30	30	2.76	5.15	4.49
May	31	31	0.01	5.10	4.47
June	30	30	0.00	7.89	5.89
July	31	31	0.00	9.36	6.19
August	31	31	0.09	8.75	5.74
September	30	30	1.03	5.98	3.90
October	31	31	0.09	4.35	1.90
November	30	30	2.51	1.31	1.06
December	31	31	1.75	1.28	1.10
Total	365	365	21.61	54.11	39.00

Effluent Production =	325,000	gpd
Treatment Pond (Lined)	1.80	acres
Approx. Storage Volume =	[illegible]	ac-ft
Approx. Storage Volume =	[illegible]	gal
Pond Percolation Rate =	0.00	in/day
Existing Ponds 5 and 6	2.41	acres
Approx. Storage Volume =	[illegible]	ac-ft
Approx. Storage Volume =	[illegible]	gal
Pond Percolation Rate =	0.33	in/day
Additional Ponds	26.3	acres
Approx. Storage Volume =	[illegible]	ac-ft
Approx. Storage Volume =	[illegible]	gal
Pond Percolation Rate =	0.33	in/day

Alfalfa Area (Available) =	0	acres
Alfalfa Area =	0	acres
Alfalfa Rootzone AWHC =	3.00	inch

STORAGE POND CALCULATIONS:

Month	Effluent Produced (gal/month)	Effluent to Alfalfa (gal/month)	Effluent to Ponds (gal/month)	Surface Rainfall (gal/month)	Surface Evaporation (gal/month)	Pond Percolation (gal/month)	Monthly Available (gal/month)	Cumulative Available (gal/month)
January	10,075,000	0	10,075,000	4,258,373	403,883	7,975,304	5,954,186	11,471,474
February	9,100,000	0	9,100,000	3,065,366	838,833	7,203,500	4,123,033	15,594,507
March	10,075,000	0	10,075,000	3,753,002	2,858,247	7,975,304	2,994,451	18,588,958
April	9,750,000	0	9,750,000	2,286,597	4,266,658	7,718,036	51,903	18,640,861
May	10,075,000	0	10,075,000	8,285	4,225,234	7,975,304	-2,117,253	16,523,608
June	9,750,000	0	9,750,000	0	6,534,615	7,718,036	-4,502,651	12,020,957
July	10,075,000	0	10,075,000	0	7,756,619	7,975,304	-5,656,923	6,364,034
August	10,075,000	0	10,075,000	74,563	7,249,177	7,975,304	-5,074,918	1,289,116
September	9,750,000	0	9,750,000	853,332	4,950,152	6,942,296	-1,289,116	0
October	10,075,000	0	10,075,000	74,563	3,603,876	6,545,687	0	0 *
November	9,750,000	0	9,750,000	2,079,478	1,087,376	7,718,036	3,024,066	3,024,066
December	10,075,000	0	10,075,000	1,449,835	1,056,309	7,975,304	2,493,222	5,517,288
Total	118,625,000	0	118,625,000	17,903,394	44,830,979	91,697,415	0	*Start at 0 Stored October 1st*
ac-ft	364	0	364	55	138	281		

RECLAMATION AREA:

Month	Effluent Applied (gal/month)	Effluent Applied (in)	Effective Rainfall (in)	Fresh Irrigation (in)	Gross Crop Need (in)	Soil Moisture Start (in)	Soil Moisture End (in)	Percolation & Leaching* >3.00 in
Alfalfa			0.0 acres		Irrigation Application Efficiency =			70%
January	0	#DIV/0!	4.29	0.00	0.59	#DIV/0!	#DIV/0!	#DIV/0!
February	0	#DIV/0!	2.94	0.00	1.26	#DIV/0!	#DIV/0!	#DIV/0!
March	0	#DIV/0!	2.72	0.00	4.24	#DIV/0!	#DIV/0!	#DIV/0!
April	0	#DIV/0!	0.00	0.00	6.41	#DIV/0!	#DIV/0!	#DIV/0!
May	0	#DIV/0!	0.00	0.00	6.39	#DIV/0!	#DIV/0!	#DIV/0!
June	0	#DIV/0!	0.00	0.00	8.41	#DIV/0!	#DIV/0!	#DIV/0!
July	0	#DIV/0!	0.00	0.00	8.84	#DIV/0!	#DIV/0!	#DIV/0!
August	0	#DIV/0!	0.00	0.00	8.20	#DIV/0!	#DIV/0!	#DIV/0!
September	0	#DIV/0!	0.00	0.00	5.57	#DIV/0!	#DIV/0!	#DIV/0!
October	0	#DIV/0!	0.00	0.00	2.71	3.00	#DIV/0!	#DIV/0!
November	0	#DIV/0!	1.82	0.00	1.51	#DIV/0!	#DIV/0!	#DIV/0!
December	0	#DIV/0!	1.11	0.00	1.57	#DIV/0!	#DIV/0!	#DIV/0!
	0	#DIV/0!	12.88	0.00	55.70			#DIV/0! #DIV/0!
	0.0	ac-ft	0.0	0.0	0.0 ac-ft			#DIV/0!
Percent of Total =	#DIV/0!		#DIV/0!	#DIV/0!				

* Leaching Ratio (LR) = ECi/((5*ECe)-ECi) 20%
LR = % of total applied water to maintain the average root zone salinity at ECe while irrigating with irrigation water quality ECi.

Total Pond Storage Available (gal)	*39,800,000*
Cumulative Storage Needed (gal)	*18,640,861*
Check	*OK*

Assumptions
1. Rainfall Data per the Western Regional Climate Center.
2. Evaporation data based on 125% of Evapotranspiration (Eto), based on DWR correspondence.
3. Evapotranspiration per the CalPoly Irrigation Training & Research Center (ITRC).
4. Percolation rates are a conservative estimate based on soil types in the area.

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